arXiv:2404.08471v1 [cs.CV] 15 Feb 2024

Revisiting Feature Prediction for Learning Visual Representations from Video

Adrien Bardes[1,2,3], **Quentin Garrido**[1,4], **Jean Ponce**[3,5,6], **Xinlei Chen**[1], **Michael Rabbat**[1], **Yann LeCun**[1,5,6], **Mahmoud Assran**[1,†], **Nicolas Ballas**[1,†]

[1]FAIR at Meta, [2]Inria, [3]École normale supérieure, CNRS, PSL Research University, [4]Univ. Gustave Eiffel, CNRS, LIGM, [5]Courant Institute, New York University, [6]Center for Data Science, New York University
[†]Joint last author

This paper explores feature prediction as a stand-alone objective for unsupervised learning from video and introduces V-JEPA, a collection of vision models trained solely using a feature prediction objective, without the use of pretrained image encoders, text, negative examples, reconstruction, or other sources of supervision. The models are trained on 2 million videos collected from public datasets and are evaluated on downstream image and video tasks. Our results show that learning by predicting video features leads to versatile visual representations that perform well on both motion and appearance-based tasks, without adaption of the model's parameters; e.g., using a frozen backbone. Our largest model, a ViT-H/16 trained only on videos, obtains 81.9% on Kinetics-400, 72.2% on Something-Something-v2, and 77.9% on ImageNet1K.

Date: April 15, 2024
Correspondence: {abardes, massran, ballasn}@meta.com
Code: https://github.com/facebookresearch/jepa
Blogpost: Click here

1 Introduction

Humans possess the remarkable ability to map low-level signals originating from the retina into a semantic spatio-temporal understanding of the world; synthesizing notions such as objects and global motion (Spelke et al., 1995). A long-standing goal of the machine learning community is to identify the principles or objectives that may guide such unsupervised learning in humans (Field, 1994; Berkes and Wiskott, 2005; Hinton, 1989). One related hypothesis is based on the *predictive feature principle* (Rao and Ballard, 1999), which posits that representations of temporally adjacent sensory stimuli should be predictive of each other.

In this work, we revisit feature prediction as a stand-alone objective for unsupervised learning of visual representations from video. Numerous advances in the field — such as the standard use of transformer architectures in vision (Dosovitskiy et al., 2020), the maturing of masked autoencoding frameworks (Xie et al., 2021; Bao et al., 2021; He et al., 2021), query-based feature pooling (Chen et al., 2022), joint-embedding predictive architectures (JEPA) (LeCun, 2022; Assran et al., 2023; Baevski et al., 2022b), and larger datasets — form a unique arsenal of tools, which we integrate in a modern and conceptually simple method, the *video joint-embedding predictive architecture* or V-JEPA, which is based solely on feature prediction, without using pretrained image encoders, text, negative examples, human annotations, or pixel-level reconstruction.

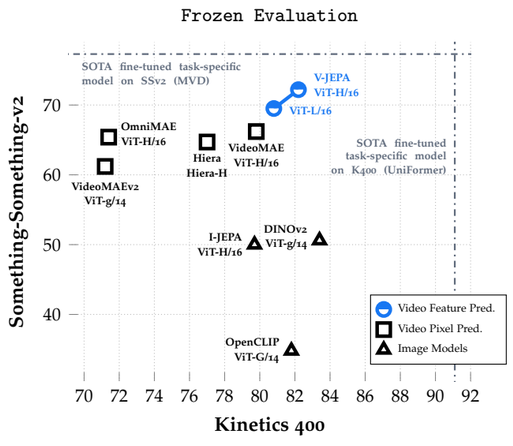


Figure 1 V-JEPA models pretrained on video learn versatile visual representations. It performs well on motion-based tasks (Something-Something-v2) and appearance-based tasks (Kinetics 400) without adaptation of the model's parameters, i.e., using the same frozen backbone for both tasks.

We seek to answer the simple question:

> *How effective is feature prediction as a stand-alone objective for unsupervised learning from video with modern tools?*

arXiv:2404.08471v1 [cs.CV] 15 Feb 2024

重新审视通过特征预测学习视频视觉表征

Adrien Bardes1,2,3, Quentin Garrido1,4, Jean Ponce3,5,6, Xinlei Chen1, Michael Rabbat1, Yann LeCun1,5,6, Mahmoud Assran[1,†]**, Nicolas Ballas**[1,†]

[1]Meta FAIR， [2]Inria， [3]巴黎高等师范学院、CNRS、PSL 研究大学， [4]古斯塔夫·埃菲尔大学、CNRS、LIGM， [5]纽约大学库朗研究所， [6]纽约大学
[†]共同末位作者

本文探索将特征预测作为视频无监督学习的独立目标，提出 V-JEPA 系列视觉模型。它们仅以特征预测训练，不使用预训练图像编码器、文本、负样本、重建或其他监督来源。模型在公共数据集收集的 200 万视频上训练，在下游图像和视频任务评估。结果表明，预测视频特征能学到通用视觉表征，无需调整模型参数（如冻结骨干），即可在运动和外观任务中表现良好。最大模型 ViT-H/16 仅用视频训练，在 Kinetics-400 达到 81.9%，Something-Something-v2 达到 72.2%，ImageNet1K 达到 77.9%。

日期：2024 年 4 月 15 日
联系：{abardes, massran, ballasn}@meta.com 代码：
https://github.com/facebookresearch/jepa 博客：点击此处

1 引言

人类能将视网膜低层信号映射为世界的语义时空理解，形成物体和全局运动等概念（Spelke 等，1995）。机器学习长期希望发现引导人类此类无监督学习的原则或目标（Field，1994；Berkes 和 Wiskott，2005；Hinton，1989）。相关假说是预测特征原则（Rao 和 Ballard，1999）：时间相邻的感官刺激表征应能互相预测。

本文重新探索特征预测作为视频无监督表征学习的独立目标。领域诸多进展，如视觉 Transformer 的标准化使用（Dosovitskiy 等，2020）、掩码自编码框架的成熟（

Xie 等，2021；Bao 等，2021；He 等，2021）、查询式特征池化（Chen 等，2022）、JEPA（LeCun，2022；Assran 等，2023；Baevski 等，2022b）和更大数据集组成独特工具组合。我们将其整合为现代、概念简单的视频联合嵌入预测架构 V-JEPA，仅使用特征预测，无需预训练图像编码器、文本、负样本、人工标注或像素

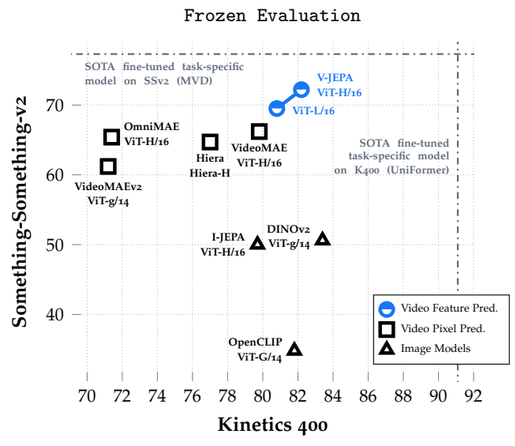


图 1. 视频预训练 V-JEPA 学到通用视觉表征。无需适配参数，用相同冻结骨干即可同时处理运动任务（SSv2）与外观任务（Kinetics 400）。

级重建。

我们希望回答一个简单问题：

> 结合现代工具，特征预测作为视频无监督学习的独立目标有多有效？

To that end, we pretrain a family of V-JEPA models on a dataset of 2 million videos collected from publicly available datasets by combining a masked modeling prediction task with a joint-embedding predictive architecture (see Figure 2). We measure performance on several downstream image and video tasks, using both frozen evaluation and end-to-end fine-tuning. Our findings suggest that feature prediction can indeed serve as an effective stand-alone objective for unsupervised learning from video, while using significantly shorter training schedules than pixel prediction methods. Specifically:

- Feature prediction leads to versatile visual representations that perform well across downstream image and video tasks without adaption of the model's weights; i.e., using a frozen backbone. V-JEPA achieves the best performance among methods we consider (+6% accuracy) on the SomethingSomething-v2 task, which requires fine-grained temporal understanding. V-JEPA is also competitive on tasks like Kinetics400, where appearance-based features are sufficient and hence state-of-the-art image models such as DINOv2 excel (Figure 1 and Table 6).
- Models trained with feature prediction are superior to pixel prediction approaches under a frozen evaluation protocol (attentive probing) and are competitive with pixel prediction under full fine-tuning, while using significantly shorter training schedules (Tables 5 and 6).
- Models trained with feature prediction are more label-efficient than pixel prediction approaches. Decreasing the available number of labeled examples results in an increase in the performance gap between V-JEPA and pixel-reconstruction models (Table 7).

2 Related Works

Slow Features. One way to encourage temporally adjacent representations to be predictive of each other is to ensure that they vary slowly over time. Early works targeting predictive features encouraged representations of individual video frames to be locally temporally invariant, while preventing representation collapse by using spectral methods, as in SFA (Wiskott and Sejnowski, 2002), SSA (Kayser et al., 2001), and Simulated Fixations (Zou et al., 2012). More recently, Goroshin et al. (2015); Wang et al. (2010) train a siamese convolutional network to map the representations of two subsequent frames to the same point, while encouraging distant frames to have diverse representations via a pairwise margin loss and a triplet loss, respectively. Other works (Oord et al., 2018; Surís et al., 2021; Feichtenhofer et al., 2021) implement temporal invariance using noise-contrastive estimation (Gutmann and Hyvärinen, 2012). Our exploration in this paper goes beyond temporal invariance and explores feature prediction using masked modeling.

Predictive Features. Going beyond local invariance, a family of works trains a predictor network to map the representation of a frame or clip at one time-step to a distinct representation at another time-step. Srivastava et al. (2015); Vondrick et al. (2016); Wang et al. (2023b) train such a video feature predictor network on top of a frozen pretrained image or video encoder. Unfreezing the target feature extractor, several methods train the video encoder and the predictor network simultaneously, while preventing collapse by using a supervised action forecasting loss (Girdhar and Grauman, 2021), or by using the representations of distant clips as negative samples in a contrastive loss (Han et al., 2019, 2020; Tan et al., 2023), often focusing on small convolutional encoders (Han et al., 2019, 2020). The idea of learning a representation by predicting missing information in feature space is also core to the joint-embedding predictive architecture (JEPA) (LeCun, 2022), which combines a siamese encoder with a predictor network. JEPAs have been successfully instantiated in several modalities, such as with audio data (Baevski et al., 2022b) and image data (Zhou et al., 2021; Oquab et al., 2023; Assran et al., 2023). In this work, we extend this paradigm to video data by leveraging recent advances in self-supervised learning.

Advances in Self-Supervised Learning. The use of vision transformers (Dosovitskiy et al., 2020; Li et al., 2022) has become standard practice in self-supervised learning with joint-embedding architectures (Chen et al., 2021; Caron et al., 2021; Oquab et al., 2023; Zhou et al., 2021; Assran et al., 2022), and unlocked masked image modeling in pixel space by parameterizing the pixel decoder as a transformer with learnable mask tokens (Dosovitskiy et al., 2020; Xie et al., 2021; He et al., 2021; Bao et al., 2021), demonstrating a step-change in the representation quality of autoencoding methods (Vincent et al., 2010). This line of generative methods was subsequently extended to video data using spatio-temporal masking (Tong et al., 2022; Feichtenhofer et al., 2022; Wang et al., 2023a; Kalluri et al., 2023; Gupta et al., 2023). It was also recently shown that the representations of masked image autoencoders could be significantly improved by using learnable pooling mechanisms based on cross-attention (Chen et al., 2022). Finally, through careful selection of design choices, the non-contrastive collapse prevention strategy in BYOL (Grill et al., 2020) was recently made to work with image feature prediction methods (Baevski et al., 2022b; Assran et al., 2023), which demonstrated the ability to learn representations that can be leveraged for various downstream tasks without relying on invariance to hand-crafted image transformations.

为此，在公共数据收集的 200 万视频上，将掩码预测任务与 JEPA 结合，预训练 V-JEPA 系列（图 2）。在多个图像、视频任务做冻结评估及端到端微调。特征预测确可独立支持有效的视频无监督学习，训练日程显著短于像素预测。具体而言：

- 特征预测学到通用视觉表征，无需调整模型权重，即冻结骨干，在图像和视频任务均表现良好。V-JEPA 在需要细粒度时间理解的 SSv2 中取得所比较方法最佳（+6%）。在外观特征已足够、DINOv2 等图像模型擅长的 K400，也具有竞争力（图 1、表 6）。
- 特征预测在冻结注意力评估中优于像素方法，在完整微调中同样有竞争力，而训练日程显著更短（表 5、6）。
- 特征预测比像素预测更高效地利用标签。标注越少，V-JEPA 与像素重建模型的性能差距越大（表 7）。

2 相关工作

慢特征。让时间相邻表征可以相互预测的一种方法，是使它们随时间缓慢变化。早期工作鼓励单帧表征局部时间不变，用谱方法防坍塌，如 SFA（Wiskott 和 Sejnowski，2002）、SSA（Kayser 等，2001）、模拟注视（Zou 等，2012）。Goroshin 等（2015）与 Wang 等（2010）训练孪生卷积网络，把相邻帧映射到同一点，分别用成对间隔损失、三元组损失让远处帧表征多样化。其他工作（Oord 等，2018；Surís 等，2021；Feichtenhofer 等，2021）用噪声对比估计（Gutmann 和 Hyvärinen，2012）实现时间不变性。本文的探索超越时间不变性，探索通过掩码建模实现特征预测。

预测特征。超越局部不变性，一类工作训练预测器，将某时刻帧或片段的表征映射为另一时刻的不同表征。Srivastava 等（2015）、Vondrick 等（2016）、Wang 等（2023b）在冻结的预训练图像或视频编码器之上训练此类预测器。另一些方法解冻目标特征提取器，同时训练视频编码器与预测器，用监督动作预测损失（Girdhar 和 Grauman，2021），或将远处片段表征作为对比负样本（Han 等，2019、2020；Tan 等，2023）防坍塌，通常关注小型卷积编码器（Han 等，2019、2020）。在特征空间预测缺失信息来学习表征，也是联合嵌入预测架构 JEPA（LeCun，2022）的核心：结合孪生编码器与预测器。JEPA 已在音频（Baevski 等，2022b）和图像（Zhou 等，2021；Oquab 等，2023；Assran 等，2023）等模态成功实例化。本文利用近期自监督进展，将该范式扩展至视频。

自监督学习进展。ViT（Dosovitskiy 等，2020；Li 等，2022）已成为联合嵌入自监督学习的标准（Chen 等，2021；Caron 等，2021；Oquab 等，2023；Zhou 等，2021；Assran 等，2022）。将像素解码器设为带可学习掩码令牌的 Transformer，也促成了像素空间掩码图像建模（Dosovitskiy 等，2020；Xie 等，2021；He 等，2021；Bao 等，2021），使自编码表征质量显著跃升（Vincent 等，2010）。这些生成方法后来通过时空遮蔽扩展到视频（Tong 等，2022；Feichtenhofer 等，2022；Wang 等，2023a；Kalluri 等，2023；Gupta 等，2023）。基于交叉注意力的可学习池化也可显著改善掩码图像自编码器表征（Chen 等，2022）。最后，仔细选择设计后，BYOL 的非对比防坍塌策略（Grill 等，2020）成功用于图像特征预测（Baevski 等，2022b；Assran 等，2023），无需对手工图像变换保持不变，就能学习用于多种下游任务的表征。

Feature Prediction versus Pixel Reconstruction. Approaches that predict in pixel space must dedicate significant model capacity and compute to capture all the low-level detail in the visual input. By contrast, approaches that predict in latent space have the flexibility to eliminate irrelevant or unpredictable pixel-level details from the target representation (Vondrick et al., 2016). Predicting in representation space has been shown to lead to versatile representations that perform well across many downstream tasks through linear probing or low-shot adaptation (Assran et al., 2023; Oquab et al., 2023; Assran et al., 2022), while demonstrating an efficiency gain during pretraining compared to pixel level reconstruction (Assran et al., 2023; Baevski et al., 2022b,a). The works of Baevski et al. (2022a,b) additionally show that predicting in representation space results in competitive end-to-end fine-tuning performance in the image, audio and text domains. In this work, we extend these findings to the video modality.

3 Methodology: Video-JEPA

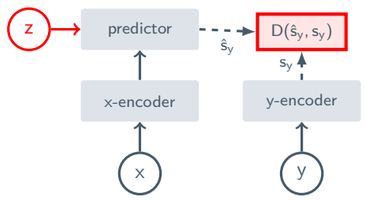


Figure 2 Joint-Embedding Predictive Architectures are trained to predict the representation of an input y from the representation of another input x. The additional variable z provides the predictor with information about the transformation that computes y from x.

Our goal is to explore the effectiveness of feature prediction as a stand-alone objective for learning visual representations from video. To that end, we use a joint-embedding predictive architecture (JEPA) (LeCun, 2022); see Figure 2. The main idea behind a JEPA is to learn by predicting the representation of an input y from the representation of another input x. The basic architecture is made up of an encoder, $E_\theta(\cdot)$, which computes the representation of the inputs, and a predictor, $P_\phi(\cdot)$, which predicts the representation of y from the representation of x, conditioned on a variable z indicating the transformation (or corruption) between x and y. Conditioning on z enables the generation of distinct predictions for various transformations of x.

3.1 Training Objective

We train our visual encoder $E_\theta(\cdot)$ to satisfy the constraint that representations computed from one part of the video, y, should be predictable from representations computed from another part of the video, x. The predictor network $P_\phi(\cdot)$, which maps the representation of x to the representation of y, is trained simultaneously with the encoder, and is provided specification of the spatio-temporal positions of y through the conditioning variable $z \leftarrow \Delta_y$.

Naively implementing the objective using the regression

$$\text{minimize}_{\theta,\phi} \quad \|P_\phi(E_\theta(x), \Delta_y) - E_\theta(y)\|_1,$$

would admit a trivial solution, where the encoder outputs a constant representation, regardless of its input. In practice, we use the following modified objective to prevent representation collapse,

$$\text{minimize}_{\theta,\phi} \quad \|P_\phi(E_\theta(x), \Delta_y) - \text{sg}(\overline{E}_\theta(y))\|_1, \quad (1)$$

where $\text{sg}(\cdot)$ denotes a stop-gradient operation, which does not backpropagate through its argument, and $\overline{E}_\theta(\cdot)$ is an exponential moving average of the network $E_\theta(\cdot)$. The use of an exponential-moving average feature extractor along with a stop-gradient and a predictor has been used as a collapse prevention strategy for image pretraining (Grill et al., 2020), and studied empirically (Xie et al., 2021) and theoretically (Tian et al., 2021). In fact, the objective in equation (1) is similar to the loss of Assran et al. (2023) used for image pretraining, but we modify it to use an ℓ_1 regression, which we found to be more stable.

Theoretical motivation. A theoretical motivation for the effectiveness of this collapse prevention strategy was proposed in Grill et al. (2020) for the BYOL method. We provide a simple adaptation of their analysis for our ℓ_1 loss. For ease of exposition, we will disregard the effect of the conditioning variable z and consider one dimensional representations. Denote the representation $\overline{E}_\theta(y)$ by a random variable Y. The optimal predictor under equation (1) is thus given by the following functional expression,

$$\begin{aligned} P^\star(E_\theta(x)) &= \text{argmin}_P \|P(E_\theta(x)) - Y\|_1 \\ &= \text{median}(Y|E_\theta(x)). \end{aligned}$$

Substituting this expression for the optimal predictor into the loss function and evaluating the expected gradient of the encoder gives

$$\nabla_\theta \mathbb{E}\|P^\star(E_\theta(x)) - Y\|_1 = \nabla_\theta \text{MAD}(Y|E_\theta(x)),$$

where $\text{MAD}(\cdot\,|E_\theta(x))$ is the median absolute deviation of a random variable conditioned on $E_\theta(x)$. Thus, in the case where the predictor is optimal, the encoder must learn to capture as much information about the video as possible to minimize the deviation of the target. The hypothesis is that incorporating an exponential moving average to compute the representation of y ensures that the predictor evolves faster than the encoder and remains close to optimal, thereby preventing collapse.

特征预测与像素重建。像素预测需大量容量和计算捕获输入所有低层细节；潜在空间预测则能灵活消除目标表征中无关或不可预测的像素细节（Vondrick 等，2016）。表征空间预测可通过线性探测或少样本适配，在多种下游任务中表现良好（Assran 等，2023；Oquab 等，2023；Assran 等，2022），且预训练比像素重建高效（Assran 等，2023；Baevski 等，2022b、a）。Baevski 等（2022a、b）还表明，它在图像、音频和文本领域端到端微调具有竞争力。本文将这些发现扩展至视频。

3 方法：Video-JEPA

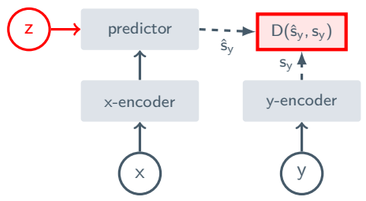


图 2. JEPA 从输入 x表征预测输入 y 表征。额外变量 z 提供从 x计算 y 的变换信息。

目标是探索特征预测作为视频表征学习独立目标的效果。采用 JEPA（LeCun，2022；图 2），从另一输入 x表征预测输入 y表征。基本架构含编码器 $E_\theta(\cdot)$计算输入表征，以及预测器$P_\phi(\cdot)$，从 x表征预测 y 表征；以表示 x 与y间变换或破坏的变量 z 为条件。条件 z 使不同 x变换获得不同预测。

3.1 训练目标

训练视觉编码器 $E_\theta(\cdot)$，使视频部分 y算得的表征，能够由另一部分的表征从视频另一部分 x计算的表征预测。将x 表征映射为 y 表征的预测器 $P_\phi(\cdot)$与编码器同时训练，并由条件变量 $z \leftarrow \Delta_y$获知 y 的时空位置。

直接以回归实现该目标，

$$\text{minimize}_{\theta,\phi} \quad \|P_\phi(E_\theta(x), \Delta_y) - E_\theta(y)\|_1,$$

会产生平凡解：编码器不论输入都输出常量。实际采用以下修改后的目标防坍塌：

$$\text{minimize}_{\theta,\phi} \quad \|P_\phi(E_\theta(x), \Delta_y) - \text{sg}(\overline{E}_\theta(y))\|_1, \quad (1)$$

其中 $\text{sg}(\cdot)$表示停止梯度，不通过参数反传； $E_\theta(\cdot)$是网络 $E_\theta(\cdot)$的指数移动平均。EMA 特征提取器结合停止梯度和预测器，曾用于图像预训练防坍塌（Grill 等，2020），并有实验（Xie 等，2021）与理论研究（Tian 等，2021）。式（1）类似 Assran 等（2023）的图像预训练损失，但改为 ℓ_1 回归，我们发现更稳定。

理论动机。Grill 等（2020）对 BYOL 防坍塌策略给出理论解释，这里简单改用于 ℓ_1损失。为便于说明，忽略条件变量 z，考虑一维表征。将 $E_\theta(y)$记为随机变量 Y，式（1）的最优预测器有如下函数表达：

$$\begin{aligned} P^\star(E_\theta(x)) &= \text{argmin}_P \|P(E_\theta(x)) - Y\|_1 \\ &= \text{median}(Y|E_\theta(x)). \end{aligned}$$

将最优预测器表达式代入损失，计算编码器的期望梯度，得到：

$$\nabla_\theta \mathbb{E}\|P^\star(E_\theta(x)) - Y\|_1 = \nabla_\theta \text{MAD}(Y|E_\theta(x)),$$

其中 $\text{MAD}(\cdot\,|E_\theta(x))$是以 $E_\theta(x)$为条件的随机变量中位数绝对偏差。因此，预测器最优时，编码器必须尽可能捕获视频信息以最小化目标偏差。假说是，用 EMA 计算 y 表征，可让预测器比编码器演化更快，保持接近最优，从而防坍塌。

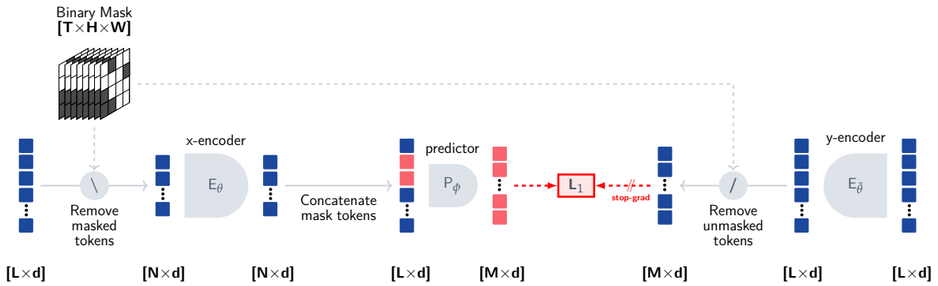


Figure 3 *V-JEPA.* Training operates on a video clip of T frames with spatial resolution $H \times W$, flattened into a sequence of L tokens. (Left to right): We first obtain the input of the x-encoder by dropping tokens from the video clip. The x-encoder then processes the masked video sequence, and outputs an embedding vector for each input token. Next, the outputs of the x-encoder are concatenated with a set of learnable mask tokens containing positional embeddings of the masked spatio-temporal patches. The predictor network processes the combined token sequence, and outputs an embedding vector for each mask token. The outputs of the predictor are then regressed to the prediction targets using an L_1 loss. The prediction targets correspond to the output of the y-encoder.

3.2 Prediction Task: Predicting y from x

The feature prediction task is based on a masked modeling formulation (He et al., 2021; Tong et al., 2022); i.e., regions x and y from the video are sampled using masking. To sample y from a video, we sample several (possibly overlapping) spatially continuous blocks with various aspect ratios and repeat the spatial blocks across the entire temporal dimension of the video; x is taken to be the complement. Masking a large continuous block that covers the full temporal dimension limits information leakage due to the spatial and temporal redundancy of videos, and results in a harder prediction task (Tong et al., 2022).

We leverage two types of masks: short-range masks, where we take the union of 8 randomly sampled target blocks covering 15% of each frame, and long-range masks, where we take the union of 2 randomly sampled target blocks covering 70% of each frame. In both cases, the aspect ratio for all sampled blocks is randomly chosen in the range $(0.75, 1.5)$. Given that both short-range and long-range masks are produced by sampling many blocks and taking their union, the result is an average masking ratio of $\sim 90\%$. We refer to our masking strategy as multi-block, and compare it to other possible masking strategies in Section 4.

3.3 Network Parameterization

We use a Vision Transformer (ViT) (Dosovitskiy et al., 2020; Arnab et al., 2021) as our video backbone. To process a video with a transformer network, we split the video clip into a 3D grid of L spatio-temporal patches, where a patch consists of a 16×16 pixel block spanning 2 consecutive frames; we refer to these spatio-temporal patches as tokens. This sequence of tokens is then directly processed by the stack of transformer blocks. Inputs x and y correspond to masked regions of a video, we apply the video masks by simply dropping a subset of the tokens. We apply masking at the input of the x-encoder, and at the output of the y-encoder to construct contextualized targets (Baevski et al., 2022b). The encoder is parameterized using standard ViT networks, while the predictor is a narrow transformer implemented using 12 blocks with an embedding dimension of 384. Taking inspiration from masked autoencoders (He et al., 2021), our predictor takes as input the sequence of embeddings produced by the x-encoder as well as a sequence of learnable mask tokens with positional embeddings indicating the spatio-temporal positions of the y tokens. The output of the predictor is an embedding vector for each mask token; see Figure 3 and refer to Appendix B for more details.

3.4 Pretraining Data and Evaluation Setup

Pretraining. We combine several public datasets to construct an unsupervised video pretraining dataset, which we refer to as VideoMix2M. Specifically, we combine the videos from HowTo100M (HT) (Miech et al., 2019), Kinetics-400/600/700 (K710) (Kay et al., 2017), and Something-Something-v2 (SSv2) (Goyal et al., 2017), and remove any overlap with the validation sets of Kinetics-400/600/700 and Something-Something-v2, resulting in approximately 2 million videos. We train a ViT-L/16, a ViT-H/16, and a ViT-H/16_{384} transformer model on VideoMix2M. We use a batch size of 3072 for the ViT-L/16 and ViT-H/16 models, and a batch size of 2400 for the ViT-H/16_{384} model. Each model takes as input a video clip of 16 frames sampled with a frame-skip of 4, corresponding to roughly 3 second clips on average. The ViT-L/16 and ViT-H/16 process the video at a spatial resolution of 224, while the ViT-H/16_{384} uses an input resolution of 384; cf. Appendix C.

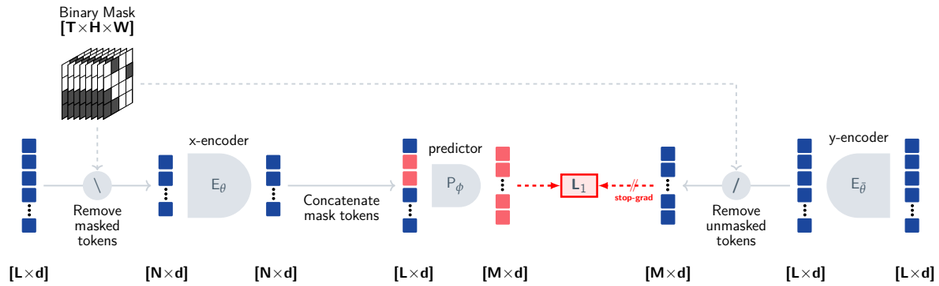


图 3. V-JEPA 训练： T帧、空间分辨率 $H \times W$的视频展平为 L 令牌序列。从左至右：先丢弃部分令牌，得到 x编码器输入；x 编码器处理遮蔽序列，为每个输入令牌输出嵌入。将 x编码器输出与可学习掩码令牌拼接，后者含遮蔽时空图块的位置嵌入。预测器处理完整序列，为每个掩码令牌输出嵌入，用 L_1 损失回归预测目标；目标来自 y编码器输出。

3.2 预测任务：从 x预测 y

特征预测采用掩码建模（He 等，2021；Tong 等，2022），通过遮蔽抽取视频区域 x 和 y 。抽取 y 时，采样多个可能重叠、纵横比不同的空间连续块，将它们贯穿完整时间轴； x 为补集。遮蔽跨完整时间的大连续块，可限制时空冗余引起的信息泄漏，使任务更难（Tong 等，2022）。

采用两种掩码：短范围，8 个随机目标块的并集，每块覆盖一帧 15%；长范围，2 个随机目标块的并集，每块覆盖 70%。所有块纵横比随机取（0.75, 1.5）。多块并集产生平均 $\sim$ 90% 掩码比例，称多块策略，第 4 节与其他策略比较。

3.3 网络参数化

视频骨干使用 ViT（Dosovitskiy 等，2020；Arnab 等，2021）。将视频划分为 L 个时空图块的三维网格，每块为跨连续 2 帧的 16×16 像素块，称令牌。令牌序列直接通过 Transformer 堆叠。输入 x 和 y 对应视频的遮蔽区域，直接丢弃部分令牌即可实施掩码。在 x编码器输入处遮蔽，在 y编码器输出处遮蔽以构造上下文化目标（Baevski 等，2022b）。编码器为标准 ViT，预测器为 12 块、384 维窄 Transformer。受 MAE（He 等，2021）启发，预测器输入包括 x编码器嵌入序列，以及带位置嵌入的可学习掩码令牌，标记 y 令牌的时空位置。输出为每个掩码令牌的嵌入向量；见图 3、附录 B。

3.4 预训练数据与评估设置

预训练。将多个公共数据集合为无监督视频集 VideoMix2M：HowTo100M（HT；Miech 等，2019）、Kinetics-400/600/700（K710；Kay 等，2017）和 SSv2（Goyal 等，2017），移除与各验证集重叠的内容，得到约 200 万视频。在该集上训练 ViT-L/16、ViT-H/16、ViT-H/16_{384} 。前两者批量 3072，ViT-H/16_{384} 批量 2400。输入为跳帧间隔 4 的 16 帧片段，平均约 3 秒。ViT-L/16、ViT-H/16 空间分辨率 224，ViT-H/16_{384}为 384；见附录 C。

Table 1 *Pixels vs. Featurized Targets.* We ablate the effect of computing the prediction loss in feature space vs pixel space. All models are trained on VideoMix2M for 90K iterations with a batch size of 3072 using the multi-block prediction task. We examine downstream performance using a frozen backbone with attentive probing, and report top-1 accuracy using a single center view. We also examine end-to-end fine-tuning performance of the models on K400. Predicting in feature space provide a consistent improvement over pixel space prediction.

		Frozen Evaluation			*Fine-Tuning*
Target	**Arch.**	**K400** (16×1×1)	**SSv2** (16×1×1)	**IN1K**	**K400-ft** (16×5×3)
Pixels	ViT-L/16	68.6	66.0	73.3	85.4
Features	ViT-L/16	**73.7**	**66.2**	**74.8**	**85.6**

Table 2 *Pretraining Data Distribution.* We pretrain all models for 90K iterations using a batch size of 3072, and evaluate downstream performance of the frozen backbones with an attentive probe using a single center view. Average performance across tasks increases with the pretraining dataset size.

			Frozen Evaluation			
Arch.	**Data**	**#Samples**	**K400** (16×1×1)	**SSv2** (16×1×1)	**IN1K**	**Avg.**
ViT-L/16	K710	700K	**75.8**	63.2	73.7	70.9
	K710+SSv2	900K	72.9	**67.4**	72.8	71.0
	K710+HT	1900K	74.5	64.2	**74.8**	71.1
	VideoMix2M	2000K	73.7	66.2	**74.8**	**71.5**
ViT-H/16	K710+SSv2	900K	**75.7**	66.8	73.7	72.0
	VideoMix2M	2000K	74.0	**68.5**	**75.9**	**72.8**

Evaluations. Pretrained models are evaluated on downstream video and image tasks. On video tasks, we use a subset of the VideoGLUE benchmark (Yuan et al., 2023) to test for various capabilities; specifically, we investigate action recognition on Kinetics-400 (K400) (Kay et al., 2017), motion classification on Something-Something-v2 (SSv2) (Goyal et al., 2017), and action localization on AVA (Gu et al., 2018). Action classification on Kinetics evaluates the appearance-based understanding of the model, as many action classes in the dataset can be inferred from the presence of specific objects in the video (Sevilla-Lara et al., 2021). Motion classification on Something-Something-v2 evaluates the temporal understanding of the model, as action classes in the dataset are decoupled from the appearance/presence of specific objects in the video (Goyal et al., 2017). Finally, action localization on AVA evaluates the ability of the model to understand and localize motions in the video. We follow standard practice and report accuracy on K400 and SSv2 by sampling several spatial and temporal views. For static image tasks, we explore object recognition on ImageNet (Russakovsky et al., 2015), scene classification on Places205 (Zhou et al., 2014), and fine-grained recognition on iNaturalist 2021 (Van Horn et al., 2018).

4 What Matters for Learning Representations from Video?

In this section we isolate the contributions of several design choices, including: a) the use of a feature prediction versus pixel prediction objective, b) the construction of the pretraining data distribution, c) the feature pooling strategy for leveraging the model's representations in downstream tasks, and d) the masking strategy, towards identifying: what to predict from what?

4.1 Predicting Representations versus Pixels

We first ablate the effect of computing the prediction loss in representation space. We train a pair of ViT-L/16 models using either a V-JEPA feature prediction loss, or a mean-squared error loss with the normalized pixel values, as in masked autoencoders (He et al., 2021), and perform a sweep over the learning rate and weight decay schedules for both approaches. All models are pretrained on VideoMix2M for 90K iterations with a batch size of 3072 using multi-block masking. We examine performance on Kinetics-400 (K400), Something-Something-v2 (SSv2), and ImageNet-1K (IN1K), using a frozen backbone with an attentive probe, and report top-1 accuracy using a single center view. We also examine end-to-end fine-tuning performance of the models on Kinetics-400.

Results of this comparison are reported in Table 1 and indicate that predicting in feature space provides a consistent performance improvement over pixel space prediction in both frozen evaluation of the video backbone, as well as end-to-end fine-tuning.

4.2 Pretraining Data Distribution

Next we study the impact of the pretraining data distribution in Table 2. Leveraging large scale datasets

表 1. 像素与特征目标。VideoMix2M、批量 3072、多块预测、90K 次训练，比较损失在特征或像素空间。冻结骨干与注意力探测器，单中心视图 top-1；另做 K400 端到端微调。特征空间预测一致优于像素空间。

		冻结评估			微调
目标	**架构**	**K400** (16×1×1)	**SSv2** (16×1×1)	**IN1K**	**K400-ft** (16×5×3)
像素	ViT-L/16	68.6	66.0	73.3	85.4
特征	ViT-L/16	**73.7**	**66.2**	**74.8**	**85.6**

表 2. 预训练数据分布。全部模型批量 3072 训练 90K 次，冻结骨干、单中心视图注意力探测。数据集越大，跨任务平均性能越好。

			冻结评估			
架构	**Data**	**样本数**	**K400** (16×1×1)	**SSv2** (16×1×1)	**IN1K**	**Avg.**
ViT-L/16	K710	700K	**75.8**	63.2	73.7	70.9
	K710+SSv2	900K	72.9	**67.4**	72.8	71.0
	K710+HT	1900K	74.5	64.2	**74.8**	71.1
	VideoMix2M	2000K	73.7	66.2	**74.8**	**71.5**
ViT-H/16	K710+SSv2	900K	**75.7**	66.8	73.7	72.0
	VideoMix2M	2000K	74.0	**68.5**	**75.9**	**72.8**

评估。下游包含视频与图像任务。视频采用 VideoGLUE（Yuan 等，2023）子集测试多种能力：Kinetics-400（K400；Kay 等，2017）动作识别，Something-Something-v2（SSv2；Goyal 等，2017）运动分类，AVA（Gu 等，2018）动作定位。Kinetics 检验外观理解，因为许多类别可由视频中特定物体的出现推断（Sevilla-Lara 等，2021）。SSv2 检验时间理解，因为动作类别与特定物体的外观／出现解耦（Goyal 等，2017）。AVA 检验理解并定位视频运动的能力。遵循标准做法，K400、SSv2 采样多个空间和时间视图并报告准确率。静态图像任务包括 ImageNet 物体识别（Russakovsky 等，2015）、Places205 场景分类（Zhou 等，2014）和 iNaturalist 2021 细粒度识别（Van Horn 等，2018）。

4 学习视频表征的关键因素

本节分别考察设计贡献，包括 a）特征预测与像素预测目标；b）预训练数据分布；c）下游利用表征的池化；d）掩码策略，以识别：从什么预测什么？

4.1 预测表征与像素

先消融在表征空间计算预测损失的影响。训练两种 ViT-L/16：V-JEPA 特征预测，或 MAE（He 等，2021）式归一化像素均方误差；均搜索学习率和权重衰减日程。在 VideoMix2M 用多块遮蔽，批量 3072，预训练 90K 次。冻结骨干，用注意力探测器，在 K400、SSv2、IN1K 单中心视图报告 top-1；另评估 K400 端到端微调。

表 1 比较显示，特征空间预测在视频骨干冻结评估和端到端微调中，都一致优于像素预测。

4.2 预训练数据分布

接着表 2 研究预训练数据分布。使用大规模数据集

Table 3 *Average Pooling vs. Adaptive Pooling.* We pool the feature map output by the frozen V-JEPA encoder using an attentive probe, which is then fed into a linear classifier for downstream supervised tasks (K400 and SSv2). We evaluate two pooling strategies: 1) average pooling (Avg.), and attentive pooling (Att.). Results are reported using a single center view. Using adaptive pooling with a cross-attention layer leads to improvements of +17.3 points on K400 and +16.1 points on SSv2.

		Frozen Evaluation			
		K400 (16×1×1)		**SSv2** (16×1×1)	
Method	**Arch.**	Avg.	Att.	Avg.	Att.
V-JEPA	ViT-L/16	56.7	**73.7**	50.1	**66.2**

has been critical for enabling the surge of advancements in other modalities, such as text and images (Kaplan et al., 2020; Cherti et al., 2023). We investigate whether a similar trend holds for video data. To control for the possible confounding variable of compute budget, we pretrain all models in Table 2 for 90K iterations using a batch-size of 3072. We report downstream results on K400, SSv2, and IN1K using a frozen backbone with an attentive probe, and report top-1 accuracy using a single center view.

Table 2 shows that average performance across tasks monotonically increases as we increase the size of the pretraining dataset, but the best task-specific performance is obtained by independently selecting the pretraining data for each specific downstream task. For instance, the L/16 obtains its best SSv2 performance when pretrained on K710+SSv2, its best K400 performance when pretrained only on K710, and its best IN1K performance when pretrained only on K710+HT. The best average performance across all tasks is achieved by pretraining VideoMix2M, which combines all the data sources. Similarly, the H/16 pretrained on K710+SSv2 achieves a greater K400 score than the H/16 pretrained on VideoMix2M, however, the top performing H/16 on average is pretrained on VideoMix2M.

4.3 Evaluation: Attentive Probing

Next we explore the feature pooling strategy for applying the model's representations in downstream tasks. Since the prediction objective in equation (1) is unnormalized, there is no a priori reason for the encoder to yield a linearly separable subspace (Chen et al., 2020). Thus, rather than using a linear operation (averaging) to pool the features output of the frozen backbone, we explore a learnable non-linear pooling strategy. Specifically, when evaluating the frozen pretrained backbone on downstream tasks, we learn a cross-attention layer with a learnable query token. The output of the cross-attention layer is then added back to the query token (residual connection), and then fed into two-layer MLP

Table 4 *Ablating Prediction Task.* Models are ViT-L/16 networks pretrained on K710 and SSv2 and evaluated with an attentive probe using a single center view. The region x is sampled by masking spatio-temporal regions in the video; y is the mask complement. **1) random-tube[r]:** x is obtained by masking a fraction r of tubes (spatial patches extended across the entire temporal duration) from the video, **2) causal multi-block[p]:** x is restricted to the first p frames of the 16-frame video, which are then masked with a random set of spatio-temporal blocks, **3) multi-block**: x is obtained by masking a random set of spatio-temporal blocks from the entire video. Best performance obtained by using multiblock masking.

	Frozen Evaluation		
Masking	**K400** (16×1×1)	**SSv2** (16×1×1)	**IN1K**
random-tube[0.9]	51.5	46.4	55.6
causal multi-block[6]	61.3	49.8	66.9
causal multi-block[12]	71.9	63.6	72.2
multi-block	**72.9**	**67.4**	**72.8**

with a single GeLU activation, followed by a LayerNorm, and finally a linear classifier.

In Table 3 we see that using adaptive pooling with a learnable cross-attention layer leads to a significant improvement of +17 points on K400 and +16.1 points on SSv2. Using an attentive-probe is also beneficial for other baseline models as reported in Appendix E.

4.4 Prediction Task: Predicting y from x

We conduct an ablation on the masking strategy used in V-JEPA pretraining. We examine the following masking strategies: `random-tube[r]` in which x is obtained by removing a random fraction r of tubes (spatial patches extended across the entire temporal duration) from the video, `causal multi-block[p]` in which x is restricted to the first p frames of the 16-frame video, which are then masked with a random set of spatio-temporal blocks, and `multi-block` in which x obtained by masking a random set of spatio-temporal blocks from the entire video. Spatio-temporal blocks are sampled using the parameters described in Section 3.2; an ablation on the size and quantity of masked spatio-temporal blocks is provided in Appendix E.4.

Table 4 indicates that the best results are obtained by sampling x using a *multi-block* strategy, wherein the network is forced to make predictions after removing large continuous blocks in the video. When x is only sampled from the first few frames of the video, as in the *causal multi-block* strategy, we observe a decrease in downstream performances. Finally, the *random-tube* strategy, wherein 90% of the tubes in the video are randomly masked, leads to features of low-semantic quality when combined with our feature prediction objective.

表 3. 平均与自适应池化。冻结 V-JEPA 输出特征图经池化后，送入线性分类器进行 K400、SSv2 监督任务。比较平均（Avg.）和注意力池化（Att.），单中心视图评估。交叉注意力自适应池化使 K400 提高 +17.3 点，SSv2 提高 +16.1 点。

		冻结评估			
		K400 (16×1×1)		**SSv2** (16×1×1)	
方法	**架构**	Avg.	Att.	Avg.	Att.
V-JEPA	ViT-L/16	56.7	**73.7**	50.1	**66.2**

对文本和图像等模态的快速进展至关重要（Kaplan 等，2020；Cherti 等，2023）。我们研究视频是否也如此。为控制计算预算，表 2 全部模型用批量 3072 训练 90K 次。冻结骨干、注意力探测器在 K400、SSv2、IN1K 单中心视图报告 top-1。

表 2 显示，预训练数据集越大，跨任务平均性能单调上升，但各任务最佳值来自单独选择相应预训练数据。例如 L/16 在 K710+SSv2 预训练时 SSv2 最佳，仅 K710 时 K400 最佳，仅 K710+HT 时 IN1K 最佳。合并全部来源的 VideoMix2M 平均性能最佳。同样，K710+SSv2 预训练 H/16 的 K400 分数高于 VideoMix2M，但平均最好的 H/16 来自 VideoMix2M。

4.3 评估：注意力探测

接着探索下游特征池化。式（1）目标未归一化，因此没有先验理由保证编码器产生线性可分子空间（Chen 等，2020）。不用平均等线性运算，而探索可学习非线性池化：评估冻结骨干时，学习带可学习查询令牌的交叉注意力层，将输出加回查询令牌（残差），再送入两层 MLP，

表 4. 预测任务消融。ViT-L/16 在 K710、SSv2 预训练，单中心视图注意力探测。 x 为遮蔽时空区域， y 为其补集。1）随机管状[r]：移除 r 比例跨完整时间的空间图块管，得到 x；2）因果多块[p]： x 限于 16 帧的前 p 帧，再随机遮蔽时空块；3）多块：对完整视频随机遮蔽时空块得到 x 。多块策略最佳。

	冻结评估		
	K400 (16×1×1)	**SSv2** (16×1×1)	**IN1K**
遮蔽			
随机管状[0.9]	51.5	46.4	55.6
因果多块[6]	61.3	49.8	66.9
因果多块[12]	71.9	63.6	72.2
多块	**72.9**	**67.4**	**72.8**

带一次 GELU，随后 LayerNorm，最后线性分类器。

表 3 中，可学习交叉注意力自适应池化使 K400 提高 +17 点，SSv2 提高 +16.1 点。其他基线也受益于注意力探测，见附录 E。

4.4 预测任务：从 x预测 y

消融 V-JEPA 预训练掩码：随机管状[r] ，移除视频中随机 r 比例的管（贯穿全部时间的空间图块）得到 x ；因果多块[p] ， x 限于前 p 帧，再随机遮蔽时空块；多块，对完整视频随机遮蔽时空块得到 x 。时空块按第 3.2 节参数抽样，大小及数量消融见附录 E.4。

表 4 中，多块策略抽取 x 最佳：移除大连续块后迫使网络预测。因果多块仅从前几帧抽取 x ，下游性能下降。随机遮蔽 90% 管的策略，结合特征预测后学到的特征语义质量较低。

Table 5 *Comparison with Pixel Prediction Methods.* We compare V-JEPA with OmniMAE (Girdhar et al., 2023), VideoMAE (Tong et al., 2022), and Hiera (Ryali et al., 2023), which leverage a pixel-reconstruction loss. All models are trained using a ViT-L architecture or a comparable Hiera-L. We evaluate the approaches on downstream image tasks (IN1K, Places205, iNat201) and video tasks (K400, SSv2, AVA) in both frozen evaluation (with a frozen backbone), and end-to-end fine-tuning. All models are evaluated at resolution 224. On K400 and SSv2 we follow the standard practice of reporting accuracy from several spatial and temporal views from the video. In frozen evaluation, V-JEPA outperforms the baselines on all downstream tasks, except ImageNet, where the model achieves 74.8% compared to 75.1% of an OmniMAE model trained directly on ImageNet. V-JEPA also achieves the best fine-tuning performance amongs all ViT-L models and matches the Hiera-L on SSv2. The V-JEPA results are achieved while processing significantly fewer examples during pretraining.

				Frozen Evaluation w/ Att. Pooling						*Fine-Tuning*	
Method	**Arch.**	**#Samples Seen**	**Iter.**	**K400** (16×8×3)	**SSv2** (16×2×3)	**AVA**	**IN1K**	**Places205**	**iNat21**	**K400-ft** (16×5×3)	**SSv2-ft** (16×2×3)
Methods pretrained using pixel prediction											
OmniMAE	ViT-L/16	2400M	1170K	65.6	60.6	14.4	**75.1**	59.8	66.1	84.0	74.2
VideoMAE	ViT-L/16	410M	400K	77.8	65.5	21.6	71.1	59.3	64.6	85.4	74.3
Hiera	Hiera-L	770M	1500K	75.5	64.2	15.8	68.9	58.5	56.9	**87.3**	**75.1**
V-JEPA	ViT-L/16	270M	90K	**80.8**	**69.5**	**25.6**	74.8	**60.3**	**67.8**	85.6	**75.1**

Table 6 *Comparison with State-of-the-Art Models.* We compare V-JEPA with state-of-the-art baselines in frozen evaluation with an attentive probe on downstream image tasks (IN1K, Place205, iNat21) and video tasks (K400, SSv2, AVA). All models are evaluated at resolution 224, except I-JEPA$_{512}$ and V-JEPA$_{384}$ which are evaluated respectively at resolution 512 and 384. On K400 and SSv2 we follow the standard practice of reporting accuracy from several spatial and temporal views from the video. Compared to other video baselines, V-JEPA exhibits a consistent improvement across all downstream tasks. Compared to image-models that excel under the frozen evaluation, V-JEPA shows a significant performance improvement on tasks requiring motion understanding (+21 points on SSv2), and reduces the gap between video and image models on tasks requiring static appearance-based features.

				Video Tasks			*Image Tasks*		
Method	**Arch.**	**Params.**	**Data**	**K400** (16×8×3)	**SSv2** (16×2×3)	**AVA**	**IN1K**	**Places205**	**iNat21**
Methods pretrained on Images									
I-JEPA	ViT-H/16_{512}	630M	IN22K	79.7	50.0	19.8	84.4	66.5	85.7
OpenCLIP	ViT-G/14	1800M	LAION	81.8	34.8	23.2	85.3	**70.2**	83.6
DINOv2	ViT-g/14	1100M	LVD-142M	**83.4**	50.6	24.3	**86.2**	68.4	**88.8**
Methods pretrained on Videos									
MVD	ViT-L/16	200M	IN1K+K400	79.4	66.5	19.7	73.3	59.4	65.7
OmniMAE	ViT-H/16	630M	IN1K+SSv2	71.4	65.4	16.0	76.3	60.6	72.4
VideoMAE	ViT-H/16	630M	K400	79.8	66.2	20.7	72.3	59.1	65.5
VideoMAEv2	ViT-g/14	1100M	Un.Hybrid	71.2	61.2	12.9	71.4	60.6	68.3
Hiera	Hiera-H	670M	K400	77.0	64.7	17.5	71.4	59.5	61.7
V-JEPA	ViT-L/16	200M	VideoMix2M	80.8	69.5	25.6	74.8	60.3	67.8
	ViT-H/16	630M		**82.0**	71.4	**25.8**	75.9	61.7	67.9
	ViT-H/16_{384}	630M		81.9	**72.2**	25.0	**77.4**	**62.8**	**72.6**

5 Comparison with Prior Work

In Section 5.1, we investigate the impact of feature prediction by comparing V-JEPA with video approaches that rely on pixel prediction, while using a similar architecture for all baselines. Subsequently, in Section 5.2, we remove the architectural constraint and report the best performance across architectures for self-supervised video and image pretraining approaches. Finally, we explore the label-efficiency of V-JEPA relative to other self-supervised video pretraining approaches in Section 5.3. We further detail the evaluation setup in Appendix D.

5.1 Comparison with Pixel Prediction

To investigate the effectiveness of feature prediction pretraining, we first compare V-JEPA to video masked modeling models relying on a pixel prediction loss. We control for the possible confounding factor of model architecture by evaluating all models using either a ViT-L/16 encoder, or a Hiera-L encoder, which has a similar number of parameters. For the pixel prediction baselines we consider VideoMAE (Tong et al., 2022; Wang et al., 2023a), which trains vision transformer autoencoders exclusively on video, Hiera (Ryali et al., 2023), which trains a hierarchical transformer autoencoder on video, and OmniMAE (Girdhar et al., 2023), which trains a vision transformer autoencoder on static images and video simultaneously.

Table 5 examines both frozen evaluation with an attentive probe on downstream video and image tasks, as well as end-to-end fine-tuning. In frozen evaluation, V-JEPA outperforms the baselines on all downstream tasks, except ImageNet, where we achieve 74.8% compared to 75.1% of an OmniMAE model trained directly on Im-

表 5. 与像素预测方法比较：OmniMAE（Girdhar 等，2023）、VideoMAE（Tong 等，2022）、Hiera（Ryali 等，2023）均使用像素重建。模型使用 ViT-L 或相近 Hiera-L。在图像（IN1K、Places205、iNat201）和视频（K400、SSv2、AVA）任务进行冻结评估及端到端微调，分辨率均为 224。K400、SSv2 采用多空间／时间视图。冻结评估时，V-JEPA 除 ImageNet 外均优于基线；ImageNet 为 74.8%，直接在其上训练的 OmniMAE 为 75.1%。V-JEPA 的 ViT-L 微调最佳，SSv2 匹配 Hiera-L，而预训练处理样本显著更少。

				注意力池化冻结评估						微调	
方法	**架构**	**样本数 Seen**	**迭代**	**K400** (16×8×3)	**SSv2** (16×2×3)	**AVA**	**IN1K**	**Places205**	**iNat21**	**K400-ft** (16×5×3)	**SSv2-ft** (16×2×3)
像素预测预训练方法											
OmniMAE	ViT-L/16	2400M	1170K	65.6	60.6	14.4	**75.1**	59.8	66.1	84.0	74.2
VideoMAE	ViT-L/16	410M	400K	77.8	65.5	21.6	71.1	59.3	64.6	85.4	74.3
Hiera	Hiera-L	770M	1500K	75.5	64.2	15.8	68.9	58.5	56.9	**87.3**	**75.1**
V-JEPA	ViT-L/16	270M	90K	**80.8**	**69.5**	**25.6**	74.8	**60.3**	**67.8**	85.6	**75.1**

表 6. 与先进模型比较。冻结骨干，以注意力探测器评估图像（IN1K、Place205、iNat21）和视频（K400、SSv2、AVA）任务。除 I-JEPA$_{512}$ 为 512、V-JEPA$_{384}$ 为 384 外，分辨率均为 224。K400、SSv2 使用多空间／时间视图。V-JEPA 相对视频基线在全部下游任务一致改善；相对擅长冻结评估的图像模型，运动理解显著提升（SSv2 高+21 个点），并缩小静态外观任务的差距。

				视频任务			图像任务		
方法	**架构**	**参数量**	**Data**	**K400** (16×8×3)	**SSv2** (16×2×3)	**AVA**	**IN1K**	**Places205**	**iNat21**
图像预训练方法									
I-JEPA	ViT-H/16_{512}	630M	IN22K	79.7	50.0	19.8	84.4	66.5	85.7
OpenCLIP	ViT-G/14	1800M	LAION	81.8	34.8	23.2	85.3	**70.2**	83.6
DINOv2	ViT-g/14	1100M	LVD-142M	**83.4**	50.6	24.3	**86.2**	68.4	**88.8**
视频预训练方法									
MVD	ViT-L/16	200M	IN1K+K400	79.4	66.5	19.7	73.3	59.4	65.7
OmniMAE	ViT-H/16	630M	IN1K+SSv2	71.4	65.4	16.0	76.3	60.6	72.4
VideoMAE	ViT-H/16	630M	K400	79.8	66.2	20.7	72.3	59.1	65.5
VideoMAEv2	ViT-g/14	1100M	无标签混合	71.2	61.2	12.9	71.4	60.6	68.3
Hiera	Hiera-H	670M	K400	77.0	64.7	17.5	71.4	59.5	61.7
V-JEPA	ViT-L/16	200M	VideoMix2M	80.8	69.5	25.6	74.8	60.3	67.8
	ViT-H/16	630M		**82.0**	71.4	**25.8**	75.9	61.7	67.9
	ViT-H/16_{384}	630M		81.9	**72.2**	25.0	**77.4**	**62.8**	**72.6**

5 与此前工作比较

第 5.1 节在相近架构下比较特征与像素预测。第 5.2 节移除架构约束，报告自监督视频和图像预训练跨架构最佳结果。第 5.3 节比较标签效率；评估细节见附录 D。

5.1 与像素预测比较

为研究特征预测预训练效果，先比较 V-JEPA 与像素预测视频掩码模型，并控制模型架构这一潜在混杂因素，通过全部使用 ViT-L/16 或参数量相似的 Hiera-L 来控制。像素基线包括只用视频训练 ViT 自编码器的 VideoMAE（Tong 等，2022；Wang 等，2023a），视频层次 Transformer 自编码器 Hiera（Ryali 等，2023），以及图像、视频联合训练 ViT 自编码器 OmniMAE（Girdhar 等，2023）。

表 5 同时考察图像、视频任务的冻结注意力探测和端到端微调。冻结评估除 ImageNet 外均超过基线；ImageNet 为 74.8%，直接在 ImageNet 上训练的 OmniMAE 为 75.1%。

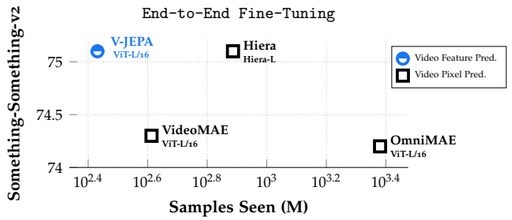


Figure 4 *SSv2 fine-tuning performance vs. Samples Seen.* We report SSv2 fine-tuning for V-JEPA and pixel-reconstruction baselines using a ViT-L/16 or Hiera-L architecture. V-JEPA outperforms all pixel-reconstruction methods using a ViT-L/16 and matches the Hiera-L performance while seeing significantly less samples during pretraining.

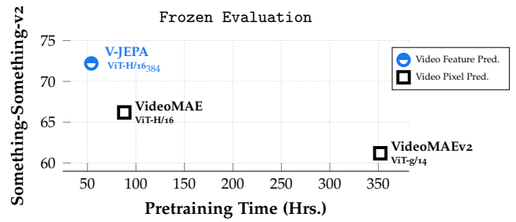


Figure 5 *SSv2 frozen-evaluation performance vs. Pretraining Time.* Wallclock times for all methods are measured on a single GPU with a batch size of 10 clips, using the official codebases for VideoMAE and VideoMAEv2, and linearly extrapolated assuming a global batch size of 2400 samples. However, note that the SSv2 accuracies of video pixel prediction methods are actually obtained with small batch sizes and significantly longer training schedules. V-JEPA outperforms pixel-reconstruction methods while training significantly faster.

ageNet; hence, V-JEPA achieves comparable ImageNet performance despite only pretraining on video.

Under the fine-tuning protocol, V-JEPA also achieves the best performance of any model trained with a ViT-L/16, and matches the performance of the Hiera-L on SSv2, which benefits from a hierachical prior (Ryali et al., 2023). The V-JEPA models achieve this result while processing significantly fewer samples during pretraining (Figure 4), demonstrating the efficiency of feature prediction as a learning principle.

5.2 Comparison with State-of-the-Art

Next, in Table 6, we inspect how the V-JEPA models pretrained on video stack up next to the largest state-of-the-art self-supervised image and video models when freezing the backbone encoder and training an attentive probe on top. Our image pretrained baselines include OpenCLIP (Cherti et al., 2023), DINOv2 (Oquab et al., 2023), and I-JEPA (Assran et al., 2023). The OpenCLIP model is trained with a contrastive image-text alignment objective, DINOv2 and I-JEPA are trained with self-supervision. These models are known to excel in their frozen-evaluation performance (Oquab et al., 2023); i.e., their ability to produce visual features that can be applied to many downstream tasks simultaneously, without end-to-end fine-tuning, and thus provide highly competitive baselines. Our video pretrained baselines include VideoMAE (Tong et al., 2022), OmniMAE (Girdhar et al., 2023), Hiera (Ryali et al., 2023), VideoMAEv2 (Wang et al., 2023a), and MVD (Wang et al., 2023b). The OpenCLIP, DINOv2 and VideoMAEv2 models are parameterized as Giant/Gigantic vision transformer architectures containing over 1B parameters trained on large-scale image or video datasets.

Comparison with video models. Compared to large-scale video baselines, the V-JEPA models outperform all previous models on every downstream video and image task with notable margin (see Table 6). Our H/16 model outperforms the largest publicly available VideoMAE, VideoMAEv2, OmniMAE, MVD, and Hiera models by at least +5 points in motion understanding (Something-Something-v2), +2 points in action recognition (Kinetics-400), +5 points on action detection (AVA), +1 point on object recognition (ImageNet-1K), +2 points in scene recognition (Places205), and +0.2 points on fine-grained recognition (iNaturalist). Moreover, when comparing pretraining wallclock time in Figure 5, we see that V-JEPA achieves this performance with a roughly 2× speedup compared to the large pixel prediction models.

Comparison with image models. On tasks that require a fine-grained understanding of motion (Something-Something-v2), the V-JEPA models provide a major improvement (over +21 points) compared to large-scale image baselines, such as DINOv2, OpenCLIP, and I-JEPA. Self-supervised pretraining from videos allows to model dynamic concepts that are not easily learned from static image datasets. Similarly, we observe that the V-JEPA models outperform image-based pretraining on action localization.

On Kinetics-400, we find image models to perform well; e.g., while DINOv2 (Oquab et al., 2023) previously reported 78.4% on K400 with a linear probe, we improve the frozen evaluation of the g/14 model to 83.4% by using an attentive probe. In this case, our H/16 model achieves 82.0% top-1 accuracy. It is worth noting that the label for many Kinetics videos can be inferred using appearance-based cues, without requiring an understanding of motion (Sevilla-Lara et al., 2021).

The V-JEPA models narrow the gap with image models on image classification tasks. In particular, V-JEPA achieves a score of 77.4% on ImageNet using a one-

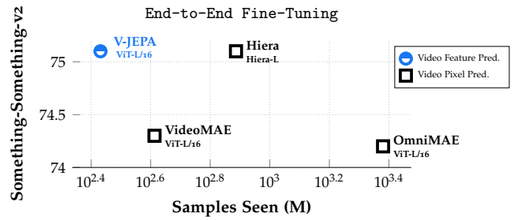


图 4. SSv2 微调性能与已见样本数。比较 ViT-L/16 或 Hiera-L 的 V-JEPA 和像素重建。V-JEPA 超过全部 ViT-L/16 像素方法，匹配 Hiera-L，而预训练样本显著更少。

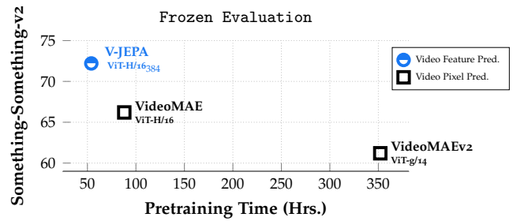


图 5. SSv2 冻结评估与预训练耗时。使用 VideoMAE、VideoMAEv2 官方代码，单 GPU、批量 10 片段测实际时间，再按全局批量 2400 线性外推。像素视频方法的 SSv2 准确率实际来自较小批量、明显更长日程。V-JEPA 训练快得多，且超过像素重建。

因此，尽管只以视频预训练，V-JEPA 的 ImageNet 性能也相近。

微调时，V-JEPA 在所有 ViT-L/16 模型中最佳，SSv2 匹配具有层次先验的 Hiera-L（Ryali 等，2023）。预训练样本显著更少（图 4），说明特征预测作为学习原则的效率。

5.2 与先进方法比较

表 6 进一步在冻结骨干、训练注意力探测器时，将视频预训练 V-JEPA 与最大规模先进自监督图像和视频模型比较。图像基线包括 OpenCLIP（Cherti 等，2023）、DINOv2（Oquab 等，2023）、I-JEPA（Assran 等，2023）。OpenCLIP 使用图文对比对齐目标，另两者自监督训练。这些模型擅长冻结评估（Oquab 等，2023），即无需端到端微调就能将视觉特征同时用于多种下游任务，因此是强基线。视频基线包括 VideoMAE（Tong 等，2022）、OmniMAE（Girdhar 等，2023）、Hiera（Ryali 等，2023）、VideoMAEv2（Wang 等，2023a）和 MVD（Wang 等，2023b）。OpenCLIP、DINOv2 和 VideoMAEv2 使用超过十亿参数的 Giant/Gigantic ViT，在大规模图像或视频数据上训练。

和图像任务都明显领先（表 6）。H/16 相对最大公开 VideoMAE、VideoMAEv2、OmniMAE、MVD、Hiera，运动理解（SSv2）至少高 +5 点，动作识别（K400）高 +2 点，动作检测（AVA）高 +5点，物体识别（IN1K）高+1 点，场景识别（Places205）高 +2 点，细粒度识别（iNaturalist）高 +0.2点。图 5 实际耗时表明，相对大型像素预测模型，达到此性能还约提速 2×。

与图像模型比较。细粒度运动理解（SSv2）中，V-JEPA 相对 DINOv2、OpenCLIP、I-JEPA 等大图像基线提高超过 +21 点。视频自监督可建模静态图像不易学到的动态概念，动作定位也优于图像预训练。

图像模型在 K400 表现良好：DINOv2（Oquab 等，2023）此前线性探测 78.4%，注意力探测将 g/14 冻结评估提升至 83.4%。此处 H/16 的 V-JEPA 为 82.0% top-1。许多 Kinetics 标签仅需外观线索即可推断，无需运动理解（Sevilla-Lara 等，2021）。

与视频模型比较。V-JEPA 相比大型视频基线，在每项下游视频

V-JEPA 缩小视频与图像分类模型差距，在 ImageNet 上使用单

Table 7 *Low-Shot Frozen Evaluation.* Comparing V-JEPA to other video models in frozen evaluation on Kinetics-400 and Something-Something-v2 as we vary the percentage of labeled examples from each dataset available for training the attentive probe. We train the probes in several low-shot settings: using either 5% of the train set, 10%, or 50%, and take 3 random splits in each setting to obtain more robust metrics, resulting in 9 different evaluation experiments for each model. We report the mean performances and standard deviation using the K400 and SSv2 validation sets. V-JEPA is more label-efficient than other models; specifically, decreasing the available number of labeled examples from each class increases the performance gap between V-JEPA and the baselines.

		Frozen Evaluation					
		K400 (16×8×3)			**SSv2** (16×2×3)		
Method	**Arch.**	5% (∼29 samples per class)	10% (∼58 samples per class)	50% (∼287 samples per class)	5% (∼48 samples per class)	10% (∼96 samples per class)	50% (∼440 samples per class)
MVD	ViT-L/16	62.6 ± 0.2	68.3 ± 0.2	77.2 ± 0.3	42.9 ± 0.8	49.5 ± 0.6	61.0 ± 0.2
VideoMAE	ViT-H/16	62.3 ± 0.3	68.5 ± 0.2	78.2 ± 0.1	41.4 ± 0.8	48.1 ± 0.2	60.5 ± 0.4
VideoMAEv2	ViT-g/14	37.0 ± 0.3	48.8 ± 0.4	67.8 ± 0.1	28.0 ± 1.0	37.3 ± 0.3	54.0 ± 0.3
V-JEPA	ViT-H/16	67.0 ± 0.2	72.1 ± 0.1	80.2 ± 0.2	51.9 ± 0.3	57.5 ± 0.4	67.3 ± 0.2
	ViT-H/16_{384}	**68.2 ± 0.2**	**72.8 ± 0.2**	**80.6 ± 0.2**	**54.0 ± 0.2**	**59.3 ± 0.5**	**67.9 ± 0.2**

layer attentive probe, which can be further improved to **77.9**% using a two-layer attentive probe. More generally, we hypothesize that the datasets used to train V-JEPA and other video models are too constrained and lack the visual diversity of the internet-scale pretraining data used by the images models; as such, there is value in focusing future work on building diverse publicly available video datasets.

5.3 Label-efficiency

We examine the label-efficiency of V-JEPA compared to other self-supervised video models by measuring the ability of the pretrained backbones to adapt to downstream tasks with few labels. Specifically, we investigate the performance of the frozen models on Kinetics-400 and Something-Something-v2 as we vary the percentage of labeled examples from each dataset available for training the attentive probe. We train the probes in several low-shot settings: using either 5% of the train set, 10%, or 50%, and take 3 random splits in each setting to obtain more robust metrics, resulting in 9 different evaluation experiments for each model. Table 7 reports the mean performances and standard deviation using the K400 and SSv2 validation sets.

We find V-JEPA to be more label-efficient than other self-supervised video models: decreasing the available number of labeled examples for training the attentive probe results in an increase in the performance gap between V-JEPA and the other models. In particular, the performance of the largest V-JEPA model on K400 drops by 12% to 68.2% top-1 when we reduce the number of labeled examples by a factor of 10× (from roughly 287 examples per class to 29 examples per class). By contrast, VideoMAEv2 drops by 30% to 37.0% top-1, VideoMAE drops by 15.9% to 62.3% top-1, and MVD drops by 14.6% to 62.6% top-1.

Similar observations hold on SSv2. The performance of the largest V-JEPA model on SSv2 drops by 13.9% to 54.0% top-1 when we reduce the number of labeled examples by a factor of 10× (from roughly 440 examples per class to 48 examples per class). By contrast, VideoMAEv2 drops by 26% to 28.0% top-1, VideoMAE drops by 19.1% to 41.4% top-1, and MVD drops by 18.1% to 42.9% top-1.

6 Evaluating the Predictor

Next, we seek to qualitatively inspect the V-JEPA models. Recall that the predictor network in V-JEPA predicts the representations of a masked spatio-temporal region y from a visible region x, given the positional information of the masked regions (see Section 3). To qualitatively investigate the grounding of the feature-space predictions, we freeze the pretrained encoder and predictor networks and train a conditional diffusion decoder to map the V-JEPA predictions to interpretable pixels. Notably, the decoder is only fed the representations predicted for the missing regions of the video, and does not have access to the unmasked regions of the video (see Figure 6a).

Given a masked video, we use the V-JEPA pretrained models to predict the representations of the missing regions, and then use the decoder to project the representations to pixel space. Figure 6b shows decoder outputs for various random seeds. Qualities that are common across samples represent information that is contained in the predictor representation.

Figure 6b shows that the V-JEPA feature predictions are indeed grounded, and exhibit spatio-temporal consistency with the unmasked regions of the video. Specifically, the samples in Figure 6b show that the V-JEPA predictor correctly captures positional uncertainty and produces a variety of visual objects at various locations with consistent motion. Some of the samples also demonstrate an understanding of object-permanence, as the visual objects remain consistent after partial occlusion.

表 7. 少样本冻结评估。改变注意力探测器可用标注比例，在 K400、SSv2 比较 V-JEPA 与其他视频模型。使用训练集 5%、10 % 或 50%，每种设置随机划分 3 次，获得稳健指标，每模型共 9 个实验。报告验证集平均性能与标准差。V-JEPA 标签效率更高：每类标注越少，领先基线的差距越大。

		冻结评估					
		K400 (16×8×3)			**SSv2** (16×2×3)		
方法	**架构**	5%（每类∼29 个样本）	10%（每类∼58 个样本）	50%（每类∼287 个样本）	5%（每类∼48 个样本）	10%（每类∼96 个样本）	50%（每类∼440 个样本）
MVD	ViT-L/16	62.6 ± 0.2	68.3 ± 0.2	77.2 ± 0.3	42.9 ± 0.8	49.5 ± 0.6	61.0 ± 0.2
VideoMAE	ViT-H/16	62.3 ± 0.3	68.5 ± 0.2	78.2 ± 0.1	41.4 ± 0.8	48.1 ± 0.2	60.5 ± 0.4
VideoMAEv2	ViT-g/14	37.0 ± 0.3	48.8 ± 0.4	67.8 ± 0.1	28.0 ± 1.0	37.3 ± 0.3	54.0 ± 0.3
V-JEPA	ViT-H/16	67.0 ± 0.2	72.1 ± 0.1	80.2 ± 0.2	51.9 ± 0.3	57.5 ± 0.4	67.3 ± 0.2
	ViT-H/16_{384}	**68.2 ± 0.2**	**72.8 ± 0.2**	**80.6 ± 0.2**	**54.0 ± 0.2**	**59.3 ± 0.5**	**67.9 ± 0.2**

层注意力探测；使用两层可进一步升至 77.9%。更一般地，我们推测视频模型训练集受限，缺少图像模型互联网规模数据的视觉多样性，未来构建更多样的公开视频数据集很有价值。

5.3 标签效率

通过少标签下的下游适配能力，比较 V-JEPA 与其他自监督视频模型的标签效率。在 K400、SSv2 冻结评估中，改变训练注意力探测器可用标注比例：训练集 5%、10%、50%，每种随机划分 3 次，每模型共 9 个实验。表 7 报告验证集平均性能与标准差。

V-JEPA 比其他视频自监督模型更高效地利用标签：注意力探测器的标注样本越少，领先差距越大。最大 V-JEPA 在 K400 将标注量减少 10× 倍（每类约 287 降至 29）后，top-1 下降 12% 至 68.2%。VideoMAEv2 下降 30% 至 37.0%，VideoMAE 下降 15.9% 至 62.3%，MVD 下降 14.6% 至 62.6%。

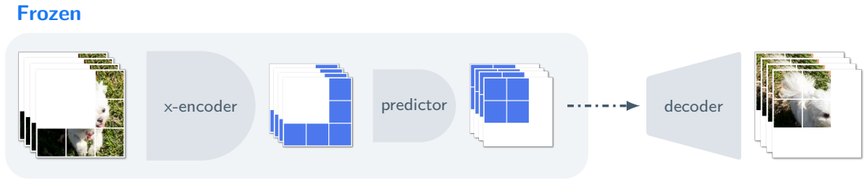
SSv2 也类似，最大 V-JEPA 下降 13.9% 至 54.0% top-1，此时标注减少 10× 倍（每类约 440 降至 48）。相对地，VideoMAEv2 下降 26% 至 28.0%，VideoMAE 下降 19.1% 至 41.4%，MVD 下降 18.1% 至 42.9%。

6 评估预测器

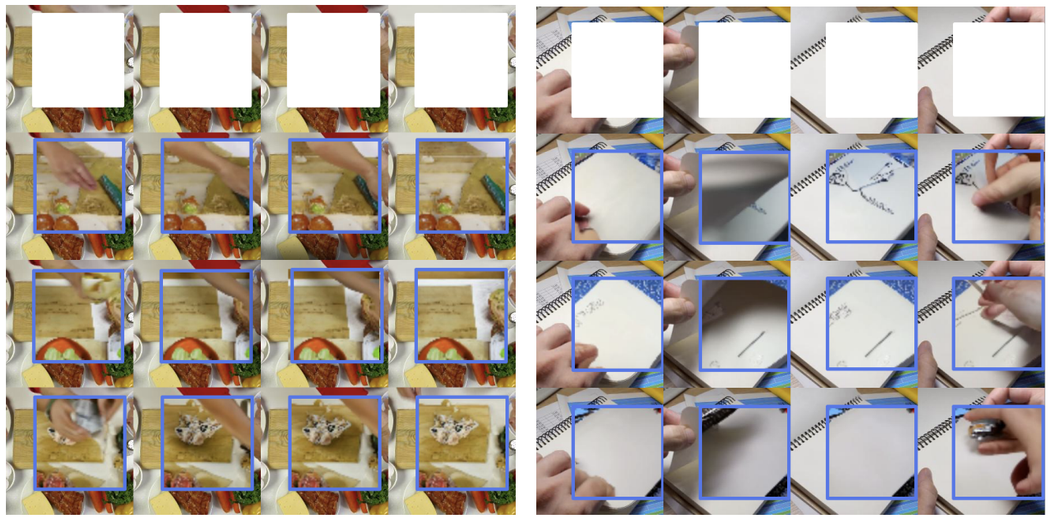
接着定性检查 V-JEPA。预测器根据遮蔽位置，从可见区域 x预测缺失时空区域 y的表征（第 3 节）。为考察特征预测是否与真实内容对应，冻结预训练编码器和预测器，训练条件扩散解码器，将预测映射为可解释像素。解码器仅接收缺失区域预测表征，无法访问未遮蔽区域（图 6a）。

给定遮蔽视频，预训练 V-JEPA 预测缺失表征，解码器投影至像素空间。图 6b 展示不同随机种子输出；共有属性代表预测表征所含信息。

图 6b 表明特征预测确与真实内容对应，与视频未遮蔽区域时空一致。预测器正确捕获位置不确定性，在不同位置产生多种视觉物体且运动连贯。部分样本还显示物体恒存理解：局部遮挡后物体仍保持一致。



(a) Visualization Methodology. We train a conditional diffusion model to decode the V-JEPA feature-space predictions to interpretable pixels; the pretrained V-JEPA encoder and predictor networks are kept frozen in this process. The decoder is only fed the representations predicted for the missing regions of the video, and does not have access to the unmasked regions of the video.



(b) Visualizations. *First Row:* Masked videos used as input to the V-JEPA models (a pretrained ViT-H/16 encoder and its corresponding predictor network). *Other rows:* Bounding boxes contain various samples from the decoder overlayed on the original video. V-JEPA is not a generative model and the decoder does not have access to the context (first row), so we do not expect samples to exactly match the input. This experiment qualitatively illustrates what information is encoded and predicted by V-JEPA. In particular, characteristics that are common across samples represent information that is encoded in the V-JEPA predictions. V-JEPA generates predictions that are spatially and temporally coherent with unmask region of the video. The predictions also capture consistent motion through time.

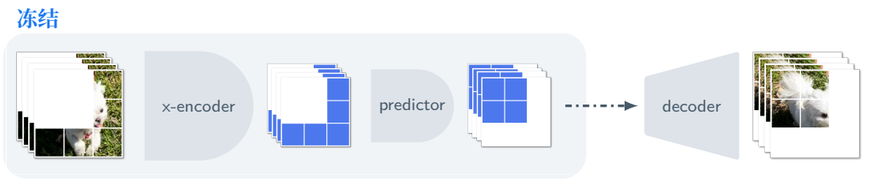
Figure 6 *Qualitative Analysis.* Offline visualizations of the V-JEPA feature-space predictions.

7 Conclusion

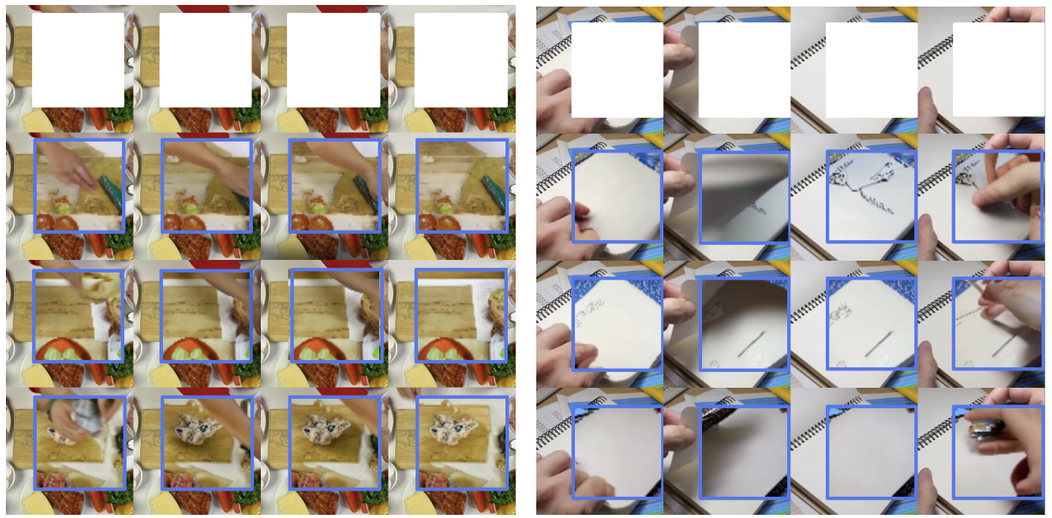
In this work, we explored the effectiveness of feature prediction as a stand-alone objective for unsupervised learning from video and introduced V-JEPA, a collection of vision models trained solely using a self-supervised feature prediction objective. The V-JEPA models demonstrate the ability to solve various downstream image and video tasks without adaption of the model parameters, and outperform previous video representation learning approaches in frozen evaluation on action recognition, spatio-temporal action detection, and image classification tasks. Additionally, we show that pretraining V-JEPA on videos is particularly effective for solving downstream tasks requiring fine-grained motion understanding, while large-scale image models trained on internet scale datasets fall short on such tasks. Finally, we empirically observed that V-JEPA models are label-efficient learners, and exhibit good performance on downstream tasks, even when only few labeled examples are available.

References

Hassan Akbari, Liangzhe Yuan, Rui Qian, Wei-Hong Chuang, Shih-Fu Chang, Yin Cui, and Boqing Gong. Vatt: Transformers for multimodal self-supervised learning from raw video, audio and text. *Advances in Neural Information Processing Systems*, 34:24206–24221, 2021.



（a）可视化方法。训练条件扩散模型，将 V-JEPA 特征预测解码为可解释像素，编码器与预测器保持冻结。解码器只接收缺失区域的预测表征，无法访问未遮蔽区域。



（b）可视化。首行：作为 V-JEPA 输入的遮蔽视频（预训练 ViT-H/16 及对应预测器）。其他行：框内为解码器样本，叠加到原视频。V-JEPA 非生成模型，解码器也无法访问首行上下文，因此不期望样本精确匹配输入。实验定性展示编码和预测的信息。样本共有属性代表预测所含信息：与未遮蔽区域空间、时间协调，并捕获时间上连续的运动。

图 6. 定性分析。V-JEPA 特征空间预测的离线可视化。

7 结论

本文探索特征预测作为视频无监督学习独立目标，提出仅用自监督特征预测训练的 V-JEPA。无需调整模型参数即可解决多种图像与视频任务，冻结评估中的动作识别、时空动作检测和图像分类优于此前视频表征方法。此外，视频预训练尤其有效地解决下游的细粒度运动理解任务，大型互联网图像模型则不擅长这些任务。实验还发现，V-JEPA 标签效率较高，即使标注很少，下游表现也良好。

参考文献

Hassan Akbari, Liangzhe Yuan, Rui Qian, Wei-Hong Chuang, Shih-Fu Chang, Yin Cui, and Boqing Gong. Vatt: Trans- formers for multimodal self-supervised learning from raw video, audio and text. Advances in Neural Information Processing Systems, 34:24206–24221, 2021.

Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lucic, and Cordelia Schmid. Vivit: A video vision transformer. In *Proceedings of the IEEE international conference on computer vision*, 2021.

Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bojanowski, Florian Bordes, Pascal Vincent, Armand Joulin, Michael Rabbat, and Nicolas Ballas. Masked siamese networks for label-efficient learning. *arXiv preprint arXiv:2204.07141*, 2022.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15619–15629, 2023.

Alexei Baevski, Arun Babu, Wei-Ning Hsu, and Michael Auli. Efficient self-supervised learning with contextualized target representations for vision, speech and language. *arXiv preprint arXiv:2212.07525*, 2022a.

Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. Data2vec: A general framework for self-supervised learning in speech, vision and language. *arXiv preprint arXiv:2202.03555*, 2022b.

Hangbo Bao, Li Dong, and Furu Wei. Beit: Bert pre-training of image transformers. *arXiv preprint arXiv:2106.08254*, 2021.

Pietro Berkes and Laurenz Wiskott. Slow feature analysis yields a rich repertoire of complex cell properties. *Journal of vision*, 5(6):9–9, 2005.

Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. Unsupervised learning of visual features by contrasting cluster assignments. *arXiv preprint arXiv:2006.09882*, 2020.

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. *arXiv preprint arXiv:2104.14294*, 2021.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. *preprint arXiv:2002.05709*, 2020.

Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, Shumin Han, Ping Luo, Gang Zeng, and Jingdong Wang. Context autoencoder for self-supervised representation learning. *arXiv preprint arXiv:2202.03026*, 2022.

Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. *arXiv preprint arXiv:2104.02057*, 2021.

Mehdi Cherti, Romain Beaumont, Ross Wightman, Mitchell Wortsman, Gabriel Ilharco, Cade Gordon, Christoph Schuhmann, Ludwig Schmidt, and Jenia Jitsev. Reproducible scaling laws for contrastive language-image learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2818–2829, 2023.

Ekin Dogus Cubuk, Barret Zoph, Vijay Mane, Dandelion and-Vasudevan, and Quoc V. Le. Autoaugment: Learning augmentation policies from data. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2019.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.

Christoph Feichtenhofer, Haoqi Fan, Bo Xiong, Ross Girshick, and Kaiming He. A large-scale study on unsupervised spatiotemporal representation learning. *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2021.

Christoph Feichtenhofer, Yanghao Li, Kaiming He, et al. Masked autoencoders as spatiotemporal learners. *Advances in neural information processing systems*, 35:35946–35958, 2022.

David J Field. What is the goal of sensory coding? *Neural computation*, 6(4):559–601, 1994.

Spyros Gidaris, Andrei Bursuc, Nikos Komodakis, Patrick Pérez, and Matthieu Cord. Learning representations by predicting bags of visual words. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6928–6938, 2020.

Rohit Girdhar and Kristen Grauman. Anticipative video transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 13505–13515, 2021.

Rohit Girdhar, Alaaeldin El-Nouby, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. Omnimae: Single model masked pretraining on images and videos. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10406–10417, 2023.

Ross Goroshin, Joan Bruna, Jonathan Tompson, David Eigen, and Yann LeCun. Unsupervised learning of spatiotemporally coherent metrics. In *Proceedings of the IEEE international conference on computer vision*, pages 4086–4093, 2015.

Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michalski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The" something something" video database for learning and evaluating visual common sense. In *Proceedings of the IEEE international conference on computer vision*, pages 5842–5850, 2017.

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. *arXiv preprint arXiv:2006.07733*, 2020.

Anurag Arnab, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lucic, and Cordelia Schmid. Vivit: A video vision transformer. In Proceedings of the IEEE international conference on computer vision, 2021.

Mahmoud Assran, Mathilde Caron, Ishan Misra, Piotr Bojanowski, Florian Bordes, Pascal Vincent, Armand Joulin, Michael Rabbat, and Nicolas Ballas. Masked siamese networks for label-efficientlearning. arXiv preprint arXiv:2204.07141, 2022.

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In Proceed- ings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 15619–15629, 2023.

Alexei Baevski, Arun Babu, Wei-Ning Hsu, and Michael Auli. Efficient self-supervised learning with contextualized target representations for vision, speech and language. arXiv preprint arXiv:2212.07525, 2022a.

Alexei Baevski, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. Data2vec: A general frame- work for self-supervised learning in speech, vision and language. arXiv preprint arXiv:2202.03555, 2022b.

Hangbo Bao, Li Dong, and Furu Wei. Beit: Bert pre-training of image transformers. arXiv preprint arXiv:2106 08254, 2021 . .

Pietro Berkes and Laurenz Wiskott. Slow feature analysis yields a rich repertoire of complex cell properties. Journal of vision, 5(6):9–9, 2005.

Mathilde Caron, Ishan Misra, Julien Mairal, Priya Goyal, Piotr Bojanowski, and Armand Joulin. Unsupervised learn- ing of visual features by contrasting cluster assignments. arXiv preprint arXiv:2006.09882, 2020.

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jé- gou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. arXiv preprint arXiv:2104.14294, 2021.

Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. preprint arXiv:2002.05709, 2020.

Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, Shumin Han, Ping Luo, Gang Zeng, and Jingdong Wang. Context autoencoder for self-supervised representation learning. arXiv preprint arXiv:2202.03026, 2022.

Xinlei Chen, Saining Xie, and Kaiming He. An empirical study of training self-supervised vision transformers. arXiv preprint arXiv:2104.02057, 2021.

Mehdi Cherti, Romain Beaumont, Ross Wightman, Mitchell Wortsman, Gabriel Ilharco, Cade Gordon, Christoph Schuhmann, Ludwig Schmidt, and Jenia Jitsev. Repro- ducible scaling laws for contrastive language-image learn- ing. In Proceedings of the IEEE/CVF Conference on Com- puter Vision and Pattern Recognition, pages 2818 2829, 2023 – .

Ekin Dogus Cubuk, Barret Zoph, Vijay Mane, Dandelion and-Vasudevan, and Quoc V. Le. Autoaugment: Learning augmentation policies from data. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2019.

AlexeyDosovitskiy, Lucas Beyer, AlexanderKolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Trans- formers for image recognition at scale. arXiv preprint arXiv:2010.11929, 2020.

Christoph Feichtenhofer, Haoqi Fan, Bo Xiong, Ross Girshick, and Kaiming He. A large-scale study on unsupervised spatiotemporal representation learning. Proceedings of the IEEE conference on computer vision and pattern recogni- tion, 2021.

Christoph Feichtenhofer, Yanghao Li, Kaiming He, et al. Masked autoencoders as spatiotemporal learners. Advances in neural information processing systems, 35 35958, 2022 35946: – .

David J Field. What is the goal of sensory coding? Neural computation, 6(4):559–601, 1994.

Spyros Gidaris, Andrei Bursuc, Nikos Komodakis, Patrick Pérez, and Matthieu Cord. Learning representations by predicting bags of visual words. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 6928–6938, 2020.

Rohit Girdhar and Kristen Grauman. Anticipative video transformer. In Proceedings of the IEEE/CVF interna- tional conference on computer vision, pages 13505 13515, 2021 – .

Rohit Girdhar, Alaaeldin El-Nouby, Mannat Singh, Kalyan Vasudev Alwala, Armand Joulin, and Ishan Misra. Omnimae: Single model masked pretraining on images and videos. In Proceedings of the IEEE/CVF Confer- ence on Computer Vision and Pattern Recognition, pages 10406–10417, 2023.

Ross Goroshin, Joan Bruna, Jonathan Tompson, David Eigen, and Yann LeCun. Unsupervised learning of spatiotempo- rally coherent metrics. In Proceedings of the IEEE inter- national conference on computer vision, pages 4086 4093, 2015 – .

Raghav Goyal, Samira Ebrahimi Kahou, Vincent Michal- ski, Joanna Materzynska, Susanne Westphal, Heuna Kim, Valentin Haenel, Ingo Fruend, Peter Yianilos, Moritz Mueller-Freitag, et al. The" something something" video database for learning and evaluating visual common sense. In Proceedings of the IEEE international conference on computer vision, pages 5842–5850, 2017.

Jean-Bastien Grill, Florian Strub, Florent Altché, Corentin Tallec, Pierre H Richemond, Elena Buchatskaya, Carl Do- ersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Moham- mad Gheshlaghi Azar, et al. Bootstrap your own latent: A new approach to self-supervised learning. arXiv preprint arXiv:2006.07733, 2020.

Chunhui Gu, Chen Sun, David A Ross, Carl Vondrick, Caroline Pantofaru, Yeqing Li, Sudheendra Vijayanarasimhan, George Toderici, Susanna Ricco, Rahul Sukthankar, et al. Ava: A video dataset of spatio-temporally localized atomic visual actions. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 6047–6056, 2018.

Agrim Gupta, Jiajun Wu, Jia Deng, and Li Fei-Fei. Siamese masked autoencoders. *arXiv preprint arXiv:2305.14344*, 2023.

Michael U Gutmann and Aapo Hyvärinen. Noise-contrastive estimation of unnormalized statistical models, with applications to natural image statistics. *Journal of machine learning research*, 13(2), 2012.

Tengda Han, Weidi Xie, and Andrew Zisserman. Video representation learning by dense predictive coding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, pages 0–0, 2019.

Tengda Han, Weidi Xie, and Andrew Zisserman. Memory-augmented dense predictive coding for video representation learning. In *European conference on computer vision*, pages 312–329. Springer, 2020.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. *arXiv preprint arXiv:2111.06377*, 2021.

Geoffrey E Hinton. Connectionist learning procedures. In *Machine learning*, pages 555–610. Elsevier, 1989.

Tarun Kalluri, Deepak Pathak, Manmohan Chandraker, and Du Tran. Flavr: Flow-agnostic video representations for fast frame interpolation. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 2071–2082, 2023.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, et al. The kinetics human action video dataset. *arXiv preprint arXiv:1705.06950*, 2017.

Christoph Kayser, Wolfgang Einhäuser, Olaf Dümmer, Peter König, and Konrad Körding. Extracting slow subspaces from natural videos leads to complex cells. In *Artificial Neural Networks—ICANN 2001: International Conference Vienna, Austria, August 21–25, 2001 Proceedings 11*, pages 1075–1080. Springer, 2001.

Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Learning representations for automatic colorization. 2016.

Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Colorization as a proxy task for visual understanding. 2017.

Yann LeCun. A path towards autonomous machine intelligence version 0.9. 2, 2022-06-27. 2022.

Hsin-Ying Lee, Jia-Bin Huang, Maneesh Singh, and Ming-Hsuan Yang. Unsupervised representation learning by sorting sequences. In *Proceedings of the IEEE international conference on computer vision*, pages 667–676, 2017.

Kunchang Li, Yali Wang, Peng Gao, Guanglu Song, Yu Liu, Hongsheng Li, and Yu Qiao. Uniformer: Unified transformer for efficient spatiotemporal representation learning. *arXiv preprint arXiv:2201.04676*, 2022.

Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

Antoine Miech, Dimitri Zhukov, Jean-Baptiste Alayrac, Makarand Tapaswi, Ivan Laptev, and Josef Sivic. Howto100m: Learning a text-video embedding by watching hundred million narrated video clips. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 2630–2640, 2019.

Mehdi Noroozi and Paolo Favaro. Unsupervised learning of visual representations by solving jigsaw puzzles. In *European conference on computer vision*, pages 69–84. Springer, 2016.

Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. *arXiv preprint arXiv:2304.07193*, 2023.

Nikhil Parthasarathy, SM Eslami, João Carreira, and Olivier J Hénaff. Self-supervised video pretraining yields strong image representations. *arXiv preprint arXiv:2210.06433*, 2022.

Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A Efros. Context encoders: Feature learning by inpainting. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2536–2544, 2016.

Silvia L Pintea, Jan C van Gemert, and Arnold WM Smeulders. Déja vu: Motion prediction in static images. In *Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part III 13*, pages 172–187. Springer, 2014.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PMLR, 2021.

Rajesh PN Rao and Dana H Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature neuroscience*, 2(1):79–87, 1999.

Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and

Chunhui Gu, Chen Sun, David A Ross, Carl Vondrick, Caro- line Pantofaru, Yeqing Li, Sudheendra Vijayanarasimhan, George Toderici, Susanna Ricco, Rahul Sukthankar, et al. Ava: A video dataset of spatio-temporally localized atomic visual actions. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 6047 6056, 2018 – .

Agrim Gupta, Jiajun Wu, Jia Deng, and Li Fei-Fei . Siamese masked autoencoders. arXiv preprint arXiv:2305 14344, 2023 . .

Michael U Gutmann and Aapo Hyvärinen . Noise-contrastive estimation of unnormalized statistical models, with appli- cations to natural image statistics. J ournal of machine learning research, 13(2), 2012.

Tengda Han, Weidi Xie, and Andrew Zisserman. Video representation learning by dense predictive coding. In Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops, pages 0–0, 2019. Tengda Han, Weidi Xie, and Andrew Zisserman. Memory-augmented dense predictive coding for video representation learning. In European conference on computer vision, pages 312–329. Springer, 2020. Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dol- lár, and Ross Girshick. Masked autoencoders are scalable vision learners. arXiv preprint arXiv:2111.06377, 2021. Geoffrey E Hinton. Connectionist learning procedures. In Machine learning, pages 555–610. Elsevier, 1989. Tarun Kalluri, Deepak Pathak, Manmohan Chandraker, and Du Tran. Flavr: Flow-agnostic video representations for fast frame interpolation. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision, pages 2071–2082, 2023. Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. arXiv preprint arXiv:2001 08361, 2020 . . Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Vi- ola, Tim Green, Trevor Back, Paul Natsev, et al. The kinetics human action video dataset. arXiv preprint arXiv:1705.06950, 2017. Christoph Kayser, Wolfgang Einhäuser, Olaf Dümmer, Peter König, and Konrad Körding. Extracting slow subspaces from natural videos leads to complex cells. In Artificial Neural Networks—ICANN 2001: International Conference Vienna, Austria, August 21–25, 2001 Proceedings 11, pages 1075–1080. Springer, 2001. Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Learning representations for automatic colorization. 2016. Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Colorization as a proxy task for visual understanding. 2017. Yann LeCun. A path towards autonomous machine intelli- gence version 0.9. 2, 2022-06-27. 2022.

Hsin-Ying Lee, Jia-Bin Huang, Maneesh Singh, and Ming-Hsuan Yang. Unsupervised representation learning by sorting sequences. In Proceedings of the IEEE international conference on computer vision, pages 667–676, 2017. Kunchang Li, Yali Wang, Peng Gao, Guanglu Song, Yu Liu, Hongsheng Li, and Yu Qiao. Uniformer: Unified trans- former for efficient spatiotemporal representation learning. arXiv preprint arXiv:2201.04676, 2022. Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. arXiv preprint arXiv:1711.05101, 2017. Antoine Miech, Dimitri Zhukov, Jean-Baptiste Alayrac, Makarand Tapaswi, Ivan Laptev, and Josef Sivic. Howto100m: Learning a text-video embedding by watch- ing hundred million narrated video clips. In Proceedings of the IEEE/CVF international conference on computer vision, pages 2630–2640, 2019. Mehdi Noroozi and Paolo Favaro. Unsupervised learning of visual representations by solving jigsaw puzzles. In Euro- pean conference on computer vision, pages 69–84. Springer, 2016. Aaron van den Oord, Yazhe Li, and Oriol Vinyals. Represen- tation learning with contrastive predictive coding. arXiv preprint arXiv:1807.03748, 2018. Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervi- sion. arXiv preprint arXiv:2304.07193, 2023. Nikhil Parthasarathy, SM Eslami, João Carreira, and Olivier J Hénaff. Self-supervised video pretraining yields strong image representations. arXiv preprint arXiv:2210.06433, 2022. Deepak Pathak, Philipp Krahenbuhl, Jeff Donahue, Trevor Darrell, and Alexei A Efros. Context encoders: Feature learning by inpainting. In Proceedings of the IEEE con- ference on computer vision and pattern recognition, pages 2536–2544, 2016. Silvia L Pintea, Jan C van Gemert, and Arnold WM S meul- ders. Déja vu: Motion prediction in static images. I n Computer Vision–ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6-12, 2014, Proceedings, Part III 13, pages 172–187. Springer, 2014. Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learn- ing transferable visual models from natural language su- pervision. In International conference on machine learning, pages 8748–8763. PMLR, 2021. Rajesh PN Rao and Dana H Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. Nature neuroscience, 2(1):79–87, 1999. Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, San- jeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and

Li Fei-Fei. Imagenet large scale visual recognition challenge. *International Journal of Computer Vision*, 115(3): 211–252, 2015.

Chaitanya Ryali, Yuan-Ting Hu, Daniel Bolya, Chen Wei, Haoqi Fan, Po-Yao Huang, Vaibhav Aggarwal, Arkabandhu Chowdhury, Omid Poursaeed, Judy Hoffman, et al. Hiera: A hierarchical vision transformer without the bells-and-whistles. *arXiv preprint arXiv:2306.00989*, 2023.

Laura Sevilla-Lara, Shengxin Zha, Zhicheng Yan, Vedanuj Goswami, Matt Feiszli, and Lorenzo Torresani. Only time can tell: Discovering temporal data for temporal modeling. In *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pages 535–544, 2021.

Elizabeth S Spelke, Peter Vishton, and Claes Von Hofsten. Object perception, object-directed action, and physical knowledge in infancy. 1995.

Nitish Srivastava, Elman Mansimov, and Ruslan Salakhudinov. Unsupervised learning of video representations using lstms. In *International conference on machine learning*, pages 843–852. PMLR, 2015.

Chen Sun, Austin Myers, Carl Vondrick, Kevin Murphy, and Cordelia Schmid. Videobert: A joint model for video and language representation learning. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 7464–7473, 2019.

Dídac Surís, Ruoshi Liu, and Carl Vondrick. Learning the predictability of the future. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12607–12617, 2021.

Reuben Tan, Matthias De Lange, Michael Iuzzolino, Bryan A Plummer, Kate Saenko, Karl Ridgeway, and Lorenzo Torresani. Multiscale video pretraining for long-term activity forecasting. *arXiv preprint arXiv:2307.12854*, 2023.

Antti Tarvainen and Harri Valpola. Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. *arXiv preprint arXiv:1703.01780*, 2017.

Yuandong Tian, Xinlei Chen, and Surya Ganguli. Understanding self-supervised learning dynamics without contrastive pairs. In *International Conference on Machine Learning*, pages 10268–10278. PMLR, 2021.

Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Videomae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. *Advances in neural information processing systems*, 35:10078–10093, 2022.

Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification and detection dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 8769–8778, 2018.

Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In *Proceedings of the 25th International Conference on Machine Learning*, ICML '08, page 1096–1103, 2008.

Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, Pierre-Antoine Manzagol, and Léon Bottou. Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion. *Journal of machine learning research*, 11(12), 2010.

Carl Vondrick, Hamed Pirsiavash, and Antonio Torralba. Anticipating visual representations from unlabeled video. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 98–106, 2016.

Fei Wang, Ping Li, and Arnd Christian Konig. Learning a bi-stochastic data similarity matrix. In *2010 IEEE International Conference on Data Mining*, pages 551–560. IEEE, 2010.

Limin Wang, Bingkun Huang, Zhiyu Zhao, Zhan Tong, Yinan He, Yi Wang, Yali Wang, and Yu Qiao. Videomae v2: Scaling video masked autoencoders with dual masking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 14549–14560, 2023a.

Rui Wang, Dongdong Chen, Zuxuan Wu, Yinpeng Chen, Xiyang Dai, Mengchen Liu, Lu Yuan, and Yu-Gang Jiang. Masked video distillation: Rethinking masked feature modeling for self-supervised video representation learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 6312–6322, 2023b.

Yi Wang, Kunchang Li, Yizhuo Li, Yinan He, Bingkun Huang, Zhiyu Zhao, Hongjie Zhang, Jilan Xu, Yi Liu, Zun Wang, et al. Internvideo: General video foundation models via generative and discriminative learning. *arXiv preprint arXiv:2212.03191*, 2022.

Laurenz Wiskott and Terrence J Sejnowski. Slow feature analysis: Unsupervised learning of invariances. *Neural computation*, 14(4):715–770, 2002.

Zhirong Wu, Yuanjun Xiong, Stella X Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3733–3742, 2018.

Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A simple framework for masked image modeling. *arXiv preprint arXiv:2111.09886*, 2021.

Dejing Xu, Jun Xiao, Zhou Zhao, Jian Shao, Di Xie, and Yueting Zhuang. Self-supervised spatiotemporal learning via video clip order prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10334–10343, 2019.

Hu Xu, Gargi Ghosh, Po-Yao Huang, Dmytro Okhonko, Armen Aghajanyan, Florian Metze, Luke Zettlemoyer, and Christoph Feichtenhofer. Videoclip: Contrastive pre-training for zero-shot video-text understanding. *arXiv preprint arXiv:2109.14084*, 2021.

Jiahui Yu, Zirui Wang, Vijay Vasudevan, Legg Yeung, Mojtaba Seyedhosseini, and Yonghui Wu. Coca: Contrastive captioners are image-text foundation models. *arXiv preprint arXiv:2205.01917*, 2022.

Li Fei-Fei. Imagenet large scale visual recognition challenge. International Journal of Computer Vision, 115(3): 211–252, 2015.

Chaitanya Ryali, Yuan-Ting Hu, Daniel Bolya, Chen Wei, Haoqi Fan, Po-Yao Huang, Vaibhav Aggarwal, Arka- bandhu Chowdhury, Omid Poursaeed, Judy Hoffman, et al. Hiera: A hierarchical vision transformer without the bells- and-whistles. arXiv preprint arXiv:2306.00989, 2023.

Laura Sevilla-Lara, Shengxin Zha, Zhicheng Yan, Vedanuj Goswami, Matt Feiszli, and Lorenzo Torresani. Only time can tell: Discovering temporal data for temporal modeling. In Proceedings of the IEEE/CVF winter conference on applications of computer vision, pages 535–544, 2021.

Elizabeth S Spelke, Peter Vishton, and Claes Von Hofsten. Object perception, object-directed action, and physical knowledge in infancy. 1995.

Nitish Srivastava, Elman Mansimov, and Ruslan Salakhudi- nov. Unsupervised learning of video representations using lstms. In International conference on machine learning, pages 843–852. PMLR, 2015.

Chen Sun, Austin Myers, Carl Vondrick, Kevin Murphy, and Cordelia Schmid. Videobert: A joint model for video and language representation learning. In Proceedings of the IEEE/CVF international conference on computer vision, pages 7464–7473, 2019.

Dídac Surís, Ruoshi Liu, and Carl Vondrick. Learning the predictability of the future. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 12607–12617, 2021.

Reuben Tan, Matthias De Lange, Michael Iuzzolino, Bryan A Plummer, Kate Saenko, Karl Ridgeway, and Lorenzo Torresani. Multiscale video pretraining for long-term activity forecasting. arXiv preprint arXiv:2307.12854, 2023. Antti Tarvainen and Harri Valpola. Mean teachers are bet- ter role models: Weight-averaged consistency targets im- prove semi-supervised deep learning results. arXiv preprint arXiv:1703.01780, 2017. Yuandong Tian, Xinlei Chen, and Surya Ganguli. Under- standing self-supervised learning dynamics without con- trastive pairs. In International Conference on Machine Learning, pages 10268–10278. PMLR, 2021. Zhan Tong, Yibing Song, Jue Wang, and Limin Wang. Video- mae: Masked autoencoders are data-efficient learners for self-supervised video pre-training. Advances in neural information processing systems, 35:10078–10093, 2022. Grant Van Horn, Oisin Mac Aodha, Yang Song, Yin Cui, Chen Sun, Alex Shepard, Hartwig Adam, Pietro Perona, and Serge Belongie. The inaturalist species classification a nd detection dataset. In Proceedings of the IEEE con- f erence on computer vision and pattern recognition, pages 8769–8778, 2018. Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre- Antoine Manzagol. Extracting and composing robust fea- tures with denoising autoencoders. I n Proceedings of the 25th International Conference on Machine Learning, ICML ' 08, page 1096–1103, 2008.

Pascal Vincent, Hugo Larochelle, Isabelle Lajoie, Yoshua Bengio, Pierre-Antoine Manzagol, and Léon Bottou. Stacked denoising autoencoders: Learning useful representations in a deep network with a local denoising criterion. Journal of machine learning research, 11(12), 2010.

Carl Vondrick, Hamed Pirsiavash, and Antonio Torralba. Anticipating visual representations from unlabeled video. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 98–106, 2016.

Fei Wang, Ping Li, and Arnd Christian Konig. Learning a bi-stochastic data similarity matrix. In 2010 IEEE International Conference on Data Mining, pages 551–560 IEEE, 2010.

Limin Wang, Bingkun Huang, Zhiyu Zhao, Zhan Tong, Yinan He, Yi Wang, Yali Wang, and Yu Qiao. Videomae v2: Scaling video masked autoencoders with dual masking. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 14549–14560, 2023a.

Rui Wang, Dongdong Chen, Zuxuan Wu, Yinpeng Chen, Xiyang Dai, Mengchen Liu, Lu Yuan, and Yu-Gang Jiang. Masked video distillation: Rethinking masked feature mod- eling for self-supervised video representation learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 6312–6322, 2023b.

Yi Wang, Kunchang Li, Yizhuo Li, Yinan He, Bingkun Huang, Zhiyu Zhao, Hongjie Zhang, Jilan Xu, Yi Liu, Zun Wang, et al. Internvideo: General video foundation models via generative and discriminative learning. arXiv preprint arXiv:2212.03191, 2022. Laurenz Wiskott and Terrence J Sejnowski. Slow feature analysis: Unsupervised learning of invariances. Neural computation, 14(4):715–770, 2002. Zhirong Wu, Yuanjun Xiong, Stella X Yu, and Dahua Lin. Unsupervised feature learning via non-parametric instance discrimination. In Proceedings of the IEEE conference on computer vision and pattern recognition, pages 3733 3742, 2018 – . Zhenda Xie, Zheng Zhang, Yue Cao, Yutong Lin, Jianmin Bao, Zhuliang Yao, Qi Dai, and Han Hu. Simmim: A sim- ple framework for masked image modeling. arXiv preprint arXiv:2111.09886, 2021. Dejing Xu, Jun Xiao, Zhou Zhao, Jian Shao, Di Xie, and Yueting Zhuang. Self-supervised spatiotemporal learn- ing via video clip order prediction. In Proceedings of the IEEE/CVF Conference on Co mputer Vision and Pattern Recognition, pages 10334–10343, 2019. Hu Xu, Gargi Ghosh, Po-Yao Huang, Dmytro Okhonko, Armen Aghajanyan, Florian Metze, Luke Zettlemoyer, and Christoph Feichtenhofer. Videoclip: Contrastive pre- training for zero-shot video-text understanding. arXiv preprint arXiv:2109.14084, 2021. Jiahui Yu, Zirui Wang, Vijay Vasudevan, Legg Yeung, Mo- jtaba Seyedhosseini, and Yonghui Wu. Coca: Contrastive captioners are image-text foundation models. arXiv preprint arXiv:2205.01917, 2022.

Liangzhe Yuan, Nitesh Bharadwaj Gundavarapu, Long Zhao, Hao Zhou, Yin Cui, Lu Jiang, Xuan Yang, Menglin Jia, Tobias Weyand, Luke Friedman, et al. Videoglue: Video general understanding evaluation of foundation models. *arXiv preprint arXiv:2307.03166*, 2023.

Rowan Zellers, Jiasen Lu, Ximing Lu, Youngjae Yu, Yanpeng Zhao, Mohammadreza Salehi, Aditya Kusupati, Jack Hessel, Ali Farhadi, and Yejin Choi. Merlot reserve: Neural script knowledge through vision and language and sound. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16375–16387, 2022.

Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Torralba, and Aude Oliva. Learning deep features for scene recognition using places database. In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K.Q. Weinberger, editors, *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc., 2014. https://proceedings.neurips.cc/paper/2014/file/3fe94a002317b5f9259f82690aeea4cd-Paper.pdf.

Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. Ibot: Image bert pre-training with online tokenizer. *arXiv preprint arXiv:2111.07832*, 2021.

Will Zou, Shenghuo Zhu, Kai Yu, and Andrew Ng. Deep learning of invariant features via simulated fixations in video. *Advances in neural information processing systems*, 25, 2012.

Liangzhe Yuan, Nitesh Bharadwaj Gundavarapu, Long Zhao, Hao Zhou, Yin Cui, Lu Jiang, Xuan Yang, Menglin Jia, Tobias Weyand, Luke Friedman, et al. Videoglue: Video general understanding evaluation of foundation models. arXiv preprint arXiv:2307.03166, 2023.

Rowan Zellers, Jiasen Lu, Ximing Lu, Youngjae Yu, Yanpeng Zhao, Mohammadreza Salehi, Aditya Kusupati, Jack Hes- sel, Ali Farhadi, and Yejin Choi. Merlot reserve: Neural script knowledge through vision and language and sound. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 16375–16387, 2022.

Bolei Zhou, Agata Lapedriza, Jianxiong Xiao, Antonio Tor- ralba, and Aude Oliva. Learning deep features for scene recognition using places database. In Z. Ghahramani, M. Welling, C. Cortes, N. Lawrence, and K.Q. Wein- berger, editors, Advances in Neural Information Pro- cessing Systems, volume 27. Curran Associates, Inc., 2014.
https://proceedings.neurips.cc/paper/2014/file/3fe94a002317b5f9259f82690aeea4cd-Paper.pdf.

Jinghao Zhou, Chen Wei, Huiyu Wang, Wei Shen, Cihang Xie, Alan Yuille, and Tao Kong. Ibot: Image bert pre-training with online tokenizer. arXiv preprint arXiv:2111 07832, 2021 . .

Will Zou, Shenghuo Zhu, Kai Yu, and Andrew Ng. Deep learning of invariant features via simulated fixations in video. Advances in neural information processing systems, 25, 2012.

14

Appendix

A Extended Related Works

We first review approaches for learning visual perception from static images before discussing strategies for learning from video.

Weakly-Supervised Learning from Static Images

One family of approaches for learning visual perception from static images trains a visual encoder to predict the representations of text captions often found accompanying images from the Web, as in CLIP (Radford et al., 2021) or CoCa (Yu et al., 2022). The largest open source CLIP model to date, numbering 2B parameters and trained on over 2B web-scraped images (Cherti et al., 2023), demonstrates impressive performance on a wide range of downstream image and video tasks. Notably, this is achieved using only the light-weight adaptation of task-specific heads, also referred to as frozen-evaluation, and does not require expensive end-to-end fine-tuning of the pretrained model.

Self-Supervised Learning from Static Images

Other approaches for learning from static images leverage unsupervised objectives. Initial works on self-supervised approaches are based on sparse coding or hand-crafted pretext tasks, such as colorization (Larsson et al., 2016, 2017), rotation prediction (Gidaris et al., 2020), and jigsaws (Noroozi and Favaro, 2016). More recent approaches leverage invariance-based objectives by training a visual encoder to be invariant to hand-crafted image transformations (Wu et al., 2018; Chen et al., 2020).

Another family of methods learn representations using denoising autoencoders (Vincent et al., 2008); image inpainting is one popular instantiation of this idea (Pathak et al., 2016). More recently, masked autoencoders (He et al., 2021) train an encoder-decoder transformer to predict missing pixels of a masked image. Follow-up work addresses the indeterminism of pixel reconstruction by exploring instantiations of masked image modeling in latent space (Baevski et al., 2022b; Assran et al., 2023; Baevski et al., 2022a). These approaches can be seen as applications of the predictive feature principle in the image modality.

There are also various methods that combine both masked image modeling and invariance criteria to learn visual representations from static images, such as iBOT (Zhou et al., 2021) and DINOv2 (Zhou et al., 2021; Oquab et al., 2023), the latter is currently the most competitive instantiation of self-supervised learning with static images, scaled to a model with over 1.1B parameters trained on a curated dataset of 142M images.

Weakly-Supervised Learning from Videos

One family of approaches for learning visual perception from videos relies on weakly-supervised guidance from closed captioning, often computed from an ASR transcription of audio data accompanying internet videos. For instance, VideoBERT (Sun et al., 2019; Xu et al., 2021) trains a video encoder to predict masked spans in the textual closed captions. Similarly, VideoCLIP (Xu et al., 2021) trains a video encoder to predict the representation of video captions computed by a text encoder. Follow-up work such as MERLOT (Zellers et al., 2022), VATT (Akbari et al., 2021), and InternVideo (Wang et al., 2022) extended VideoCLIP by incorporating additional unsupervised objectives.

Self-Supervised Learning from Videos

Similar to unsupervised learning from images, a family of unsupervised video representation learning approaches enforces a spatio-temporal representation of a video clip to be invariant to hand-crafted spatio-temporal data augmentations (Parthasarathy et al., 2022). However, one obvious insight is that the temporal ordering of visual information in video can provide implicit supervision. Indeed, this insight is the key insight leveraged by many works on unsupervised video learning. Towards leveraging temporal information as supervision, some approaches train a visual encoder by predicting the temporal ordering of frames (Xu et al., 2019; Lee et al., 2017). Other approaches seek to predict low-level motion vectors computed from optical flow (Pintea et al., 2014), or to predict mixing pixels in video frames, using either a frame-interpolation objective (Kalluri et al., 2023) or a denoising autoencoder (Tong et al., 2022; Feichtenhofer et al., 2022; Wang et al., 2023a).

附录

A 扩展相关工作

先回顾静态图像视觉学习，再讨论视频学习策略。

静态图像弱监督学习

一类静态图像视觉方法训练编码器，预测网络图像附带的文本描述表征，如 CLIP（Radford 等，2021）、CoCa（Yu 等，2022）。当时最大开源 CLIP 有 20 亿参数，以超过 20 亿网络图像训练（Cherti 等，2023），在广泛图像和视频任务表现突出。只需轻量适配任务头，即冻结评估，无需昂贵的端到端微调。

静态图像自监督学习

其他静态图像方法采用无监督目标。早期自监督基于稀疏编码或手工前置任务，如着色（Larsson 等，2016、2017）、旋转预测（Gidaris 等，2020）、拼图（Noroozi 和 Favaro，2016）。近期则训练视觉编码器对手工图像变换保持不变（Wu 等，2018；Chen 等，2020）。

另一类方法用去噪自编码器学习表征（Vincent 等，2008），图像补全是常见实例（Pathak 等，2016）。MAE（He 等，2021）训练编码器–解码器 Transformer，预测遮蔽图像缺失像素。后续在潜在空间探索掩码建模，处理像素重建的不确定性（Baevski 等，2022b；Assran 等，2023；Baevski 等，2022a）。这些可视为预测特征原则在图像模态的应用。

还有方法结合掩码建模与不变性准则学习静态图像表征，如 iBOT（Zhou 等，2021）和 DINOv2（Zhou 等，2021；Oquab 等，2023）。后者是当前最有竞争力的静态图像自监督实例，扩展至超过 11 亿参数，在精选的 1.42 亿图像上训练。

视频弱监督学习

一类视频视觉方法依赖字幕弱监督，字幕常由互联网视频音频的 ASR 转写得到。例如 VideoBERT（Sun 等，2019；Xu 等，2021）训练视频编码器预测字幕中的遮蔽片段；VideoCLIP（Xu 等，2021）预测文本编码器算得的字幕表征。MERLOT（Zellers 等，2022）、VATT（Akbari 等，2021）、InternVideo（Wang 等，2022）通过加入其他无监督目标扩展 VideoCLIP。

视频自监督学习

类似图像无监督学习，一类视频方法让片段的时空表征对手工时空增强保持不变（Parthasarathy 等，2022）。但视频视觉信息的时间顺序本身就可提供隐式监督，许多无监督视频工作利用此点。有些通过预测帧顺序训练视觉编码器（Xu 等，2019；Lee 等，2017）；另一些预测由光流计算的低层运动向量（Pintea 等，2014），或通过插帧目标（Kalluri 等，2023）、去噪自编码器（Tong 等，2022；Feichtenhofer 等，2022；Wang 等，2023a）预测视频帧中缺失像素。

B Extended Description of V-JEPA

In this section, we provide an in-depth description of our approach V-JEPA that is illustrated in Figure 3.

Input. Unless stated otherwise, during during pretraining, we always randomly sample a clip of 16 frames from each input video with a temporal stride of 4 between sampled frames. An input video clip therefore covers 64 frames in total, or roughly 2 seconds of a given video running at 30 frames per second. We then resize the video's spatial dimensions to 224×224, resulting in an overall shape of $16 \times 224 \times 224 \times 3$ for the entire clip. Since ViT networks process a 1D sequence of tokens, we must convert an input video clip into a 1D token sequence. To do so, we apply a 3D convolution comprising d filters of size $2 \times 16 \times 16$ with a temporal stride of 2 and a spatial stride of 16, resulting in a tensor of shape $8 \times 14 \times 14 \times d$. Next we add absolute 3D sin-cos positional embeddings to the spatio-temporal feature map and flatten it, resulting in a 1D token sequence of shape $1568 \times d$. This process is demonstrated in Figure 7.

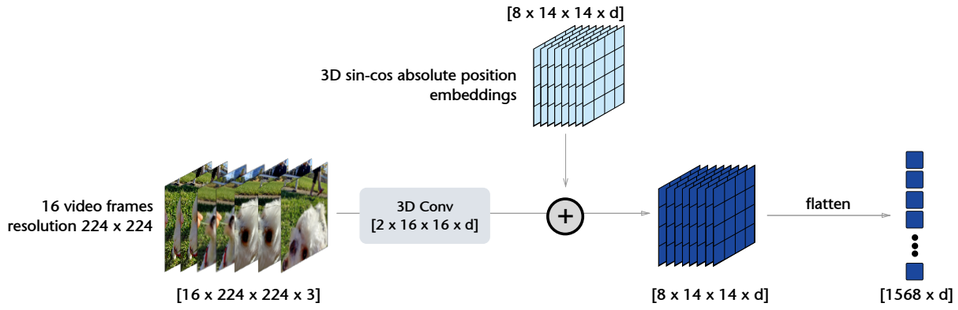


Figure 7 V-JEPA training operates on a video clip flattened into a sequence of tokens. To convert a video clip of size $16 \times 224 \times 224 \times 3$ into a 1D token sequence, we apply a 3D convolution comprising d filters of size $2 \times 16 \times 16$ with a temporal stride of 2 and a spatial stride of 16, resulting in a tensor of shape $8 \times 14 \times 14 \times d$. Next we add absolute 3D sin-cos positional embeddings to the spatio-temporal feature map and flatten it, resulting in a 1D token sequence of shape $1568 \times d$.

V-JEPA. We sample both a video clip, and a video mask in each iteration. We denote a video clip represented as a 1D token sequence of length $L = 1568$ by $x_L = (x_1, \ldots, x_L)$. Similarly, given a mask of $M < L$ patches, leaving $N = L - M$ patches unmasked, we denote the indices of masked patches by $(i_1, \ldots, i_M)$ and its complement (the indices of unmasked patches) by $(j_1, \ldots, j_N)$.

Computing the x-representations. To compute the V-JEPA loss, we first produce the x-representations by masking the video clip and feeding it into the x-encoder; we denote the masked video by $x_N = (x_{j_1}, \ldots, x_{j_N})$. Applying the x-encoder $E_\theta(\cdot)$ to the masked clip gives a sequence of patch representations, denoted as $z_N = E_\theta(x_N) = (z_{j_1}, \ldots, z_{j_N})$.

Predicting the target. Next, the V-JEPA predictor network $P_\phi(\cdot, \cdot)$ takes as input the tokens produced by the x-encoder and predicts the missing regions in the video clip, which are specified by a set of learnable mask tokens. Specifically, the mask tokens are parameterized as the sum of a shared learnable vector and an absolute 3D sin-cos positional embedding, denoted by $m_M = (m_{i_1}, \ldots, m_{i_M})$. The output of the predictor is thus given by, $\hat{s}_M = P_\phi(z_N, m_M) = (\hat{s}_{i_1}, \ldots, \hat{s}_{i_M})$, corresponding to a d-dimensional output for each of the M masked patches.

Computing the y-representations. Finally to compute the prediction targets, the entire unmasked video clip is processed by the y-encoder to obtain a set of target representations, denoted by $s_L = \overline{E}_\theta(x_L) = (s_1, \ldots, s_L)$. The V-JEPA loss is now computed as

$$\text{Loss} = \frac{1}{M} \sum_{k \in (i_1, \ldots, i_M)} \|\hat{s}_k - s_k\|_1, \tag{2}$$

which is simply the average L_1 distance between the output of the predictor and the y-encoder. We then compute a gradient update with respect to the parameters of the x-encoder, θ, and the predictor, ϕ, and subsequently update the parameters of the y-encoder as an exponential moving average of the context encoder weights (Polyak average).

B V-JEPA 的详细描述

本节深入描述图 3 的 V-JEPA。

输入。除另有说明，预训练始终随机采样 16 帧，采样帧间时间步幅 4，因此总覆盖 64 帧，相当于 30 帧／秒视频约 2 秒。空间维度缩放为 224×224，完整片段形状为 $16 \times 224 \times 224 \times 3$ 。ViT 处理一维令牌序列，需要转换视频：以 d 个大小 $2 \times 16 \times 16$ 的三维卷积滤波器，时间步幅 2、空间步幅 16，得到形状 $8 \times 14 \times 14 \times d$的张量；加入绝对三维正弦–余弦位置嵌入，再展平为形状 $1568 \times d$的一维令牌序列。图 7 展示此过程。

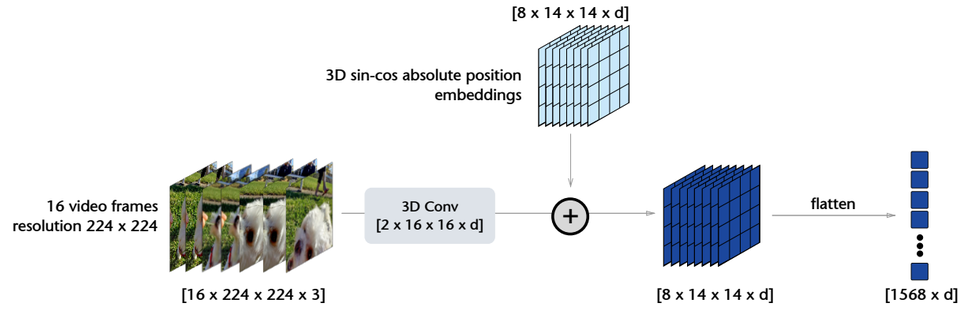


图 7. 视频展平为令牌序列。大小$16 \times 224 \times 224 \times 3$的视频经 d 个大小 $2 \times 16 \times 16$ 的三维卷积滤波器，时间步幅 2、空间步幅 16，得到形状 $8 \times 14 \times 14 \times d$张量。加入绝对三维正弦–余弦位置嵌入并展平，形成形状 $1568 \times d$的一维序列。

V-JEPA。每次迭代抽取视频与掩码。长度 $L = 1568$ 的一维视频令牌序列记为 $x_L = (x_1, \ldots, x_L)$。遮蔽 $M < L$ 个图块、保留$N = L - M$个时，遮蔽索引为$(i_1, \ldots, i_M)$，其补集，即可见索引，为$(j_1, \ldots, j_N)$。

计算 x表征。先遮蔽片段，送入 x编码器得到 x表征。遮蔽视频记为 $x_N = (x_{j_1}, \ldots, x_{j_N})$， x编码器 $E_\theta(\cdot)$输出图块表征序列 $z_N = E_\theta(x_N) = (z_{j_1}, \ldots, z_{j_N})$。

预测目标。V-JEPA 预测器 $P_\phi(\cdot, \cdot)$接收x编码器令牌，预测可学习掩码令牌指定的缺失区域。令牌为共享可学习向量加绝对三维正弦–余弦位置嵌入，记作 $m_M = (m_{i_1}, \ldots, m_{i_M})$。输出为$\hat{s}_M = P_\phi(z_N, m_M) = (\hat{s}_{i_1}, \ldots, \hat{s}_{i_M})$，对 M 个遮蔽图块各给出 d维输出。

计算 y表征。将完整未遮蔽片段送入 y编码器，得到目标序列 $s_L = E_\theta(x_L) = (s_1, \ldots, s_L)$，随后计算 V-JEPA 损失:

$$\text{Loss} = \frac{1}{M} \sum_{k \in (i_1, \ldots, i_M)} \|\hat{s}_k - s_k\|_1, \tag{2}$$

即预测器输出与 y编码器之间的平均 L_1 距离。对 x编码器参数 θ及预测器参数 ϕ做梯度更新，再用上下文编码器权重的 EMA（Polyak 平均）更新 y编码器。

Table 8 **pretraining hyper-parameters for V-JEPA.**

Hyper-parameter	ViT-L/16_{224}	ViT-H/16_{224}	ViT-H/16_{384}
data			
datasets	VideoMix2M	VideoMix2M	VideoMix2M
resolution	224	224	384
num_frames	16	16	16
temporal_stride	4	4	4
horizontal_flip	true	true	true
random_resize_scale	(0.3, 1.0)	(0.3, 1.0)	(0.3, 1.0)
random_resize_aspect_ratio	(0.75, 1.35)	(0.75, 1.35)	(0.75, 1.35)
masking			
block_aspect_ratio	(0.75, 1.5)	(0.75, 1.5)	(0.75, 1.5)
shortrange_mask_num_blocks	8	8	8
shortrange_mask_spatial_scale	0.15	0.15	0.15
longrange_mask_num_blocks	2	2	2
longrange_mask_spatial_scale	0.7	0.7	0.7
optimization			
batch_size	3072	3072	2400
total_number_of_iterations	90000	90000	90000
warmup_iterations	12000	12000	12000
lr	6.25e-4	6.25×10^{-4}	6.25×10^{-4}
start_lr	2×10^{-4}	2×10^{-4}	2×10^{-4}
final_lr	1×10^{-6}	1×10^{-6}	1×10^{-6}
start_momentum	0.998	0.998	0.998
final_momentum	1.0	1.0	1.0
start_weight_decay	0.04	0.04	0.04
final_weight_decay	0.4	0.4	0.4
scheduler_scale_factor	1.25	1.25	1.25
architecture			
patch_size	16	16	16
tubelet_size	2	2	2
pred_depth	12	12	12
pred_embed_dim	384	384	384
hardware			
dtype	bfloat16	bfloat16	bfloat16
accelerator	A100 80G	A100 80G	A100 80G

Multi-Mask Prediction. To increase the efficiency of V-JEPA, we use a multi-masking strategy (Caron et al., 2020; Baevski et al., 2022a), which enables us to amortize the cost of the target computation. As mentioned in Section 3, for a given video clip, we sample 2 different masks, short-range and long-range. While we need to forward propagate the x-encoder and predictor separately for each mask, we only need to compute the y-representation once.

C Pretraining details

In section, we report V-JEPA pretraining details. Table 8 summarizes the main hyperparameters used during pretraining.

Architectures. We use Vision Transformer (Dosovitskiy et al., 2020) (ViT) architectures for the x-encoder and y-encoder. We train three V-JEPA encoders: a ViT-L/16_{224}, a ViT-H/16_{224} and a ViT-H/16_{384}. All three encoders take as input a short video clip of 16 frames with a temporal stride of 4 between consecutive frames. The subscripts, 224 and 384, indicate the spatial resolution of the video clip. V-JEPA flattens the video clip into a sequence of non-overlapping spatio-temporal patches of size $16 \times 16 \times 2$ (see Figure 7). For all three models, the predictor is designed as a narrow ViT architecture, consisting of 12 transformer blocks with an embedding dimension of 384. For simplicity, we keep the number of self-attention heads in the predictor equal to that of the backbone used for the context-encoder/target-encoder. V-JEPA is pretrained *without* using a [cls] token.

Optimization. We use AdamW (Loshchilov and Hutter, 2017) to optimize the x-encoder and predictor weights. The ViT-L/16_{224} and ViT-H/16_{224} models use a batch size of 3072 while the ViT-H/16_{384} uses a batch size of 2400. Models are trained for a total of 90,000 iterations. The learning rate is linearly increased from 2×10^{-4} to 6.25×10^{-4} during the first $12,000$ iterations of pretraining, and decayed to 10^{-6} following a cosine schedule.

表 8. V-JEPA 预训练超参数。

超参数	ViT-L/16_{224}	ViT-H/16_{224}	ViT-H/16_{384}
data			
数据集	VideoMix2M	VideoMix2M	VideoMix2M
分辨率	224	224	384
num_帧数	16	16	16
时间_步幅	4	4	4
水平_flip	true	true	true
随机_缩放_尺度	(0.3, 1.0)	(0.3, 1.0)	(0.3, 1.0)
随机_缩放_纵横_比例	(0.75, 1.35)	(0.75, 1.35)	(0.75, 1.35)
遮蔽			
块_纵横_比例	(0.75, 1.5)	(0.75, 1.5)	(0.75, 1.5)
短范围_mask_num_块数	8	8	8
短范围_mask_空间_尺度	0.15	0.15	0.15
长范围_mask_num_块数	2	2	2
长范围_mask_空间_尺度	0.7	0.7	0.7
优化			
批量_size	3072	3072	2400
总计_数量_of_迭代次数	90000	90000	90000
预热_迭代次数	12000	12000	12000
lr	6.25e-4	6.25×10^{-4}	6.25×10^{-4}
初始_lr	2×10^{-4}	2×10^{-4}	2×10^{-4}
最终_lr	1×10^{-6}	1×10^{-6}	1×10^{-6}
初始_动量	0.998	0.998	0.998
最终_动量	1.0	1.0	1.0
初始_权重_衰减	0.04	0.04	0.04
最终_权重_衰减	0.4	0.4	0.4
调度器_尺度_系数	1.25	1.25	1.25
架构			
图块_size	16	16	16
时空图块_size	2	2	2
pred_深度	12	12	12
pred_嵌入_dim	384	384	384
硬件			
数据类型	bfloat16	bfloat16	bfloat16
加速器	A100 80G	A100 80G	A100 80G

多掩码预测。为提高效率，用多掩码（Caron 等，2020；Baevski 等，2022a）分摊目标计算成本。每片段采短范围和长范围两种掩码；各掩码需分别运行 x编码器和预测器，但 y表征只需算一次。

C 预训练细节

本节报告 V-JEPA 预训练细节，表 8 汇总主要超参数。

架构。 x编码器与y编码器采用 ViT（Dosovitskiy 等，2020）。训练三个 V-JEPA：ViT-L/16_{224}、ViT-H/16_{224}、ViT-H/16_{384}，输入均为 16 帧、时间步幅 4 的短视频。下标 224、384 表示空间分辨率。视频展平为大小 $16 \times 16 \times 2$ 的不重叠时空图块序列（图 7）。三种模型的预测器均为窄 ViT：12 个 Transformer 块，嵌入维度 384。为简单起见，预测器注意力头数与上下文／目标编码器骨干相同。预训练不使用 [cls] 令牌。

优化。AdamW（Loshchilov 和 Hutter，2017）优化 x编码器与预测器。ViT-L/16_{224}、ViT-H/16_{224} 批量 3072，ViT-H/16_{384} 批量 2400，共训练 90,000 次。前 12,000 次学习率从 2×10^{-4}线性升至 6.25×10^{-4}，随后按余弦日程降至 10^{-6}。

Table 9 Frozen Evaluation hyper-parameters.

Hyper-parameter	K400	SSv2	IN1K	Place205	iNat21
data					
num_clips	8	1	N.A.	N.A.	N.A.
num_frames	16	16	N.A.	N.A.	N.A.
temporal_stride	4	4	N.A.	N.A.	N.A.
horizontal_flip	true	true	true	true	true
random_resize_scale	(0.08, 1.0)	(0.08, 1.0)	(0.08, 1.0)	(0.08, 1.0)	(0.08, 1.0)
random_resize_aspect_ratio	(0.75, 1.33)	(0.75, 1.33)	(0.75, 1.33)	(0.75, 1.33)	(0.75, 1.33)
auto_augment	false	false	true	true	true
optimization					
batch_size	256	256	1024	1024	1024
epochs	20	20	20	20	20
lr	1e-3	1e-3	1e-3	1e-3	1e-3
final_lr	0	0	0	0	0
weight_decay	0.01	0.01	0.01	0.01	0.01

Weight-decay is also linearly increased from 0.04 to 0.4 throughout pretraining. The y-encoder weights are initialized identically to the x-encoder, and subsequently updated as an exponential moving average (EMA) (Tarvainen and Valpola, 2017) of the x-encoder weights using a momentum value which starts at 0.998 and is linearly increased to 1.0 during training (Caron et al., 2021; Assran et al., 2022). We scale all hyper-parameter schedules 25% beyond the actual training schedule. Specifically, the learning rate schedule, weight-decay schedule, and EMA schedule are computed assuming a training length of 112,500 iterations, even though we only train our model for 90,000 iterations. We found the last 25% of the default scheduler period to update hyper-parameters too aggressively, and simply truncating the schedulers improved performance.

Masking. As described in Section 3, we propose a 3D Multi-Block masking strategy. We use two type of masks: short-range masks, where we take the union of 8 randomly sampled target blocks with a spatial scale of 0.15, and long-range masks, where we take the union of 2 randomly sampled target blocks with a spatial scale of 0.7. In both cases, the aspect ratio for all sampled blocks is randomly chosen in the range $(0.75, 1.5)$.

D Evaluation details

D.1 Frozen classification

Attentive Probing. Given an input video, x_L, the V-JEPA target encoder $\overline{E}_\theta(\cdot)$ outputs a sequence of L tokens, $E_\theta(x_L) = (s_1, \ldots, s_L)$, where $s_i \in \mathbb{R}^d$. To pool this sequence of tokens into a single feature vector, we apply a lightweight non-linear cross-attention block which replace the self-attention operation of a transformer block with cross attention. Specifically, the cross-attention performs the following computation:

$$\sum_{i=1}^{L} \frac{\exp(q^\top \mathbf{W_k} s_i)}{\sum_j \exp(q^\top \mathbf{W_k} s_j)} \mathbf{W_v} s_i,$$

where $\mathbf{W_k}, \mathbf{W_v} \in \mathbf{R^{d \times d}}$ are the key and value matrices, and $q \in R^d$ is a learnable query token. The output of the cross-attention is then added back to the query token (residual connection), and then fed into two-layer MLP with a single GeLU activation, followed by a LayerNorm, and finally a linear classifier. The parameters of the cross-attention block are jointly learned with that of the linear classifier for the downstream task, while the encoder parameters are kept frozen. Note that, in practice, we actually use an attentive probe with 12 heads, each of dimension 12. In Appendix E we show that baselines benefit from the attentive probing protocol.

Optimization. For all the tasks, we use AdamW optimizer with a cosine scheduler (no warmup) that decays the learning rate from 0.001 to 0. We use a fixed weight-decay of 0.01 and apply simple data augmentations (random resized crops and horizontal flips) during training of the attentive probe, except on image tasks, where we apply AutoAugment (Dogus Cubuk et al., 2019). Table 9 reports the hyperparameters for each downstream evaluation.

Extension to multiple clips. Unless stated otherwise, our attentive probe takes 8 clips of 16 frames as input on Kinetics, and 2 clips of 16 frames on Something-Somethingv2 to increase the temporal coverage of the video.

表 9. 冻结评估超参数。

超参数	K400	SSv2	IN1K	Place205	iNat21
data					
num_片段数	8	1	N.A.	N.A.	N.A.
num_帧数	16	16	N.A.	N.A.	N.A.
时间 _步幅	4	4	N.A.	N.A.	N.A.
水平 _flip	true	true	true	true	true
随机 _缩放 _尺度	(0.08, 1.0)	(0.08, 1.0)	(0.08, 1.0)	(0.08, 1.0)	(0.08, 1.0)
随机 _缩放 _纵横 _比例	(0.75, 1.33)	(0.75, 1.33)	(0.75, 1.33)	(0.75, 1.33)	(0.75, 1.33)
auto_增强	false	false	true	true	true
优化					
批量 _size	256	256	1024	1024	1024
轮数	20	20	20	20	20
lr	1e-3	1e-3	1e-3	1e-3	1e-3
最终 _lr	0	0	0	0	0
权重 _衰减	0.01	0.01	0.01	0.01	0.01

预训练权重衰减从 0.04 线性增至 0.4。 y编码器初始化与 x编码器相同，随后以 x编码器权重的 EMA（Tarvainen 和 Valpola，2017）更新，动量从 0.998 线性增至 1.0（Caron 等，2021；Assran 等，2022）。所有超参数日程比实际训练长 25%：按 112,500 次计算学习率、衰减、EMA 日程，但仅训练 90,000 次。默认调度最后 25% 更新过于激进，直接截断可改善性能。

遮蔽。第 3 节的三维多块策略采用短范围：8 个空间尺度 0.15 的随机目标块并集；长范围：2 个尺度 0.7 的随机目标块并集。纵横比均在（0.75, 1.5）随机取。

D 评估细节

D.1 冻结分类

注意力探测。视频 x_L经 V-JEPA 目标编码器 $E_\theta(\cdot)$得到 L 个令牌$E_\theta(x_L) = (\overline{s_1}, \ldots, s_L)$，其中 $s_i \in \mathbb{R}^d$。用轻量非线性交叉注意力块将序列汇总为一个向量，即用交叉注意力替换 Transformer 的自注意力，具体计算：

$$\sum_{i=1}^{L} \frac{\exp(q^\top \mathbf{W_k} s_i)}{\sum_j \exp(q^\top \mathbf{W_k} s_j)} \mathbf{W_v} s_i,$$

其中 $\mathrm{W_k}, \mathrm{W_v} \in \mathrm{R^{d \times d}}$ 为键、值矩阵， $q \in R^d$ 为可学习查询令牌。交叉注意力输出加回查询（残差），送入带一次 GELU 的两层 MLP，随后 LayerNorm，最后线性分类器。交叉注意力与分类器参数在下游联合学习，编码器冻结。实际注意力探测器使用 12 个头，每头 12 维。附录 E 说明基线也受益于此评估协议。

优化。所有任务用 AdamW 与余弦调度（无预热），学习率 0.001 降至 0，权重衰减固定 0.01。注意力探测训练用随机缩放裁剪、水平翻转；图像任务例外，使用 AutoAugment（Dogus Cubuk 等，2019）。各评估超参数见表 9。

扩展至多片段。除另有说明，Kinetics 使用 8 个 16 帧片段，SSv2 使用 2 个，以增加时间覆盖。

Table 10 Frozen Detection hyper-parameters.

Hyper-parameter	ViT-L/16	ViT-H/16
out_layers	[18, 20, 22, 24]	[26, 28, 30, 32]
batch_size	64	64
epochs	30	30
opt	AdamW	AdamW
opt_eps	0.00000001	0.00000001
momentum	0.9	0.9
weight_decay	0.05	0.05
lr	0.0001	0.0001
warmup_lr	0.000001	0.000001
min_lr	0.000001	0.000001
warmup_epochs	2	2
warmup_steps	1	1

Specifically, we first divide a video in 8 (or 2) equal-length temporal segments, and sample 1 clip at random per segment. The video encoder $\overline{E}_\theta$ processes each clip separately and produces a clip-level feature map. The feature maps for each clip are then concatenated together and fed to the attentive probe. At test time, we average the prediction of 3 spatial views following standard practice in video classification.

Application of video models to images. To evaluate the video models on image tasks, we simply duplicate input images to generate still video clips of 16 frames. We perform this duplication operation simply for convenience in evaluation of the video models, however we find this step to be unnecessary in general. Given a video tokenizer implemented as a 3D-conv with a temporal stride of 2, it is sufficient to simply duplicate the image into a 2 frame video clip. This would result in the same number of input tokens as that produced by a static image model with a 2D-conv tokenizer.

Application of image models to videos. To evaluate image models such as DINOv2 and OpenCLIP on video tasks, we simply process each frame independently with the image encoder to produce a frame-level feature map. The feature maps for each frame are then concatenated and fed to the attentive probe, just as we do with the clip-level feature maps when evaluating video models.

D.2 Frozen detection

We evaluate our model on the AVA (Gu et al., 2018) spatio-temporal localization of human actions dataset, containing 211k training and 57k validation video segments. We follow the experimental protocol of (Feichtenhofer et al., 2021), and use precomputed masks from a pretrained Faster-RCNN adapted to videos, which uses a ResNeXt-101-FPN backbone and is pretrained on ImageNet and COCO. We train a linear classifier on top of the *frozen* V-JEPA features to classify the extracted regions of interest and report mean Average Precision (mAP) on the 60 most common classes. Hyper-parameters are provided in Table 10. Our frozen features are obtained by concatenating the last layer of the transformer encoder with three intermediate layers. We use a batch size of 64 and pretrain for 30 epochs with AdamW using a learning rate of 0.0001 with 2 epochs of warmup and a weight decay of 0.05.

D.3 Finetuning

Following Tong et al. (2022), we finetune a linear layer on top of our model, using a layer decay schema and mixup as the data augmentation pipeline. We provide all hyper-parameters for both K400 and SSv2 in Table 11.

E Extra Results

E.1 Frozen Evaluation.

Linear vs. Attentive probe Table 12 shows that V-JEPA and VideoMAE benefit from using a non-linear attentive probe and multiple clips on the K400 and SSv2 downstream tasks. Additionally, Table 13 shows that attentive probing leads to better performance on average for DINOv2 and OpenCLIP models. Since attentive probing and multiclips eval improves the performance of all models, we use it as our default protocol in frozen evaluation.

表 10. 冻结检测超参数。

超参数	ViT-L/16	ViT-H/16
out_层数	[18, 20, 22, 24]	[26, 28, 30, 32]
批量 _size	64	64
轮数	30	30
opt	AdamW	AdamW
opt_eps	0.00000001	0.00000001
动量	0.9	0.9
权重 _衰减	0.05	0.05
lr	0.0001	0.0001
预热 _lr	0.000001	0.000001
min_lr	0.000001	0.000001
预热 _轮数	2	2
预热 _步数	1	1

先将视频均分 8（或 2）段，每段随机抽 1 片段。编码器 E_θ 分别生成各片段特征图，拼接后送入注意力探测器。测试时按视频分类惯例，对 3 个空间视图的预测求平均。

视频模型应用于图像。评估时复制图像为 16 帧静态视频，主要是方便操作，通常并非必要。若令牌化器是时间步幅 2 的三维卷积，复制成 2 帧已足够；输入令牌数与使用二维卷积的静态图像模型相同。

图像模型用于视频。DINOv2、OpenCLIP 等逐帧编码，生成帧级特征图，再拼接送入注意力探测器，与视频模型的片段特征图处理相同。

D.2 冻结检测

在 AVA 人体动作时空定位集（Gu 等，2018）评估，含 211k 训练、57k 验证视频段。遵循 Feichtenhofer 等（2021），使用适配视频的预训练 Faster-RCNN 预计算掩码，其骨干 ResNeXt-101-FPN 在 ImageNet、COCO 预训练。冻结 V-JEPA 特征，在其上训练线性分类器，分类兴趣区域，报告最常见 60 类的 mAP。超参数见表 10。冻结特征拼接编码器最后一层及三个中间层。批量 64，AdamW 训练 30 轮，学习率 0.0001、预热 2 轮、权重衰减 0.05。

D.3 微调

遵循 Tong 等（2022），在模型之上微调线性层，采用逐层衰减和 mixup 增强。K400、SSv2 全部超参数见表 11。

E 额外结果

E.1 冻结评估

线性与注意力探测。表 12 显示，V-JEPA 和 VideoMAE 在 K400、SSv2 受益于非线性注意力探测和多片段。表 13 表明 DINOv2、OpenCLIP 平均也更好。由于所有模型都改善，采用多片段注意力探测作为默认冻结协议。

Table 11 Finetuning Evaluation hyper-parameters.

Hyper-parameter	K400		SSv2	
data				
num_segments	1			
num_frames	16			
sampling_rate	4			
resolution	224			
model				
model_name	ViT-L/16	ViT-H/16	ViT-L/16	ViT-H/16
drop_path	0.1	0.2	0.2	0.2
head_drop_rate	0.	0.	0.5	0.5
optimization				
batch_size	256	1024	256	256
epochs	35	25	15	15
opt	adamw			
opt_eps	0.00000001			
momentum	0.9			
weight_decay	0.05			
lr	0.002	0.0005	0.0005	0.0005
layer_decay	0.75	0.75	0.75	0.75
warmup_lr	1e-6	1e-8	1e-6	1e-6
min_lr	1e-6	1e-5	1.5e-4	1.5e-3
warmup_epochs	5			
augmentations				
color_jitter	0.4			
horizontal_flip	True	True	False	False
num_sample	2			
aa	rand-m7-n4-mstd0.5-inc1			
smoothing	0.1			
train_interpolation	bicubic			
test_num_segment	5	5	2	2
test_num_crop	3	3	3	3
erase				
prob	0.25			
mode	pixel			
count	1			
split	False			
mixup				
mixup	0.8			
cutmix	1.0			
mixup_prob	1.0			
mixup_switch_prob	0.5			
mixup_mode	batch			

表 11. 微调评估超参数。

超参数	K400		SSv2	
data				
num_分段数	1			
num_帧数	16			
采样_rate	4			
分辨率	224			
模型				
模型名_	ViT-L/16	ViT-H/16	ViT-L/16	ViT-H/16
drop_path	0.1	0.2	0.2	0.2
head_drop_rate	0.	0.	0.5	0.5
优化				
批量_size	256	1024	256	256
轮数	35	25	15	15
opt	adamw			
opt_eps	0.00000001			
动量	0.9			
权重_衰减	0.05			
lr	0.002	0.0005	0.0005	0.0005
层_衰减	0.75	0.75	0.75	0.75
预热_lr	1e-6	1e-8	1e-6	1e-6
min_lr	1e-6	1e-5	1.5e-4	1.5e-3
预热_轮数	5			
数据增强				
颜色_扰动	0.4			
水平_flip	True	True	False	False
num_样本	2			
aa	rand-m7-n4-mstd0.5-inc1			
平滑	0.1			
训练_插值	双三次			
test_num_片段	5	5	2	2
test_num_crop	3	3	3	3
擦除				
prob	0.25			
mode	像素			
数量	1			
划分	False			
mixup				
mixup	0.8			
cutmix	1.0			
mixup_prob	1.0			
mixup_开关_prob	0.5			
mixup_mode	批量			

Table 12 Linear vs. Attentive Probe Evaluation for V-JEPA and VideoMAE. We evaluate the effect of linear (Lin.) and attentive (Att.) probing when adapting V-JEPA to the K400 ($16 \times 5 \times 3$) and SSv2 ($16 \times 2 \times 2$) tasks. V-JEPA and VideoMAE benefit from using a non-linear attentive probe.

		K400		SSv2	
Method	**Arch.**	Lin.	Att.	Lin.	Att.
VideoMAE	ViT-L/16	52.5	77.8	41.3	61.2
V-JEPA	ViT-L/16	56.7	**80.8**	50.1	**69.5**

Table 13 Linear vs. Attentive Probe Evaluation for DINOv2 and OpenCLIP. We evaluate the effect of linear (Lin.) and attentive probing (Att.) when adapting DINOv2 and OpenCLIP. Image-baselines benefit from using an attentive probing strategy. Results shown in gray are reported from the linear probe evaluation in Oquab et al. (2023).

		K400		SSv2		IN1K		Place205		iNat21	
Method	**Arch.**	Lin.	Att.	Lin.	Att.	Lin.	Att.	Lin.	Att.	Lin.	Att.
DINOv2	ViT-g/14	78.4	83.4	38.3	50.0	86.5	86.2	67.5	68.4	85.7	88.8
OpenCLIP	ViT-G/14	78.3	81.8	35.8	34.8	86.2	85.3	69.8	70.2	76.0	83.6

One Clip vs Multiple clips. We examine the impact of changing the temporal coverage of a model during downstream evaluation on K400 action classification. In Table 14, we evaluate VideoMAE and V-JEPA models using an attentive probe with access to either the feature map of 1 clip randomly sampled from the video, or the concatenated feature map of 8 clips randomly sampled from the video. To sample 8 clips from a video, we first divide the video into 8 equal length temporal segments, and sample 1 clip at random from each segment. A single clip corresponds to ≈ 2 seconds of a video on average, while 8 clips correspond to ≈ 16 seconds. The video encoders processes each clip separately to produce a clip-level feature map, which are then concatenated at the input to the attentive probe.

Increasing the temporal coverage from 1 clip per video to 8 clips improves the performance of both V-JEPA and VideoMAE on K400 action classification. We therefore use the multiclip attentive probing setup as our default evaluation pipeline.

E.2 Finetuning

In Table 15, we evaluate V-JEPA using finetuning (separately) on K400 and SSv2. We compare V-JEPA with VideoMAEv2 (Wang et al., 2023a), VideoMAE (Tong et al., 2022) and MVD (Wang et al., 2023b) using a ViT-L/16 or a ViT-H/16 architecture. V-JEPA obtains competitive performance using a finetuning protocol. With a ViTiH/16 architecture, V-JEPA outperforms by 1.2% VideoMAE and +0.3% VideoMAEv2 on the SSv2 dataset, while obtaining comparable performance on K400. V-JEPA also obtains performance similar to MVD on the SSv2 dataset. The MVD model achieves the best performance across models on the K400 dataset, and is trained using the image dataset ImageNet1K, in contrast to the other methods in the table, which only use video data. Additionally MVD requires the processing of significantly more samples during pretraining due to the cost of training the teacher encoder networks in a pre-pre-training step.

E.3 Sample Efficiency of pretraining

We compare the sample efficiency of pretraining various state-of-the-art image and video models. Specifically, we look at the number of samples (image or video clips) processed by the network during pretraining, which is larger than the size of the pretraining dataset for multi-epoch training. Notably, our results with V-JEPA are obtained while processing an order of magnitude fewer samples than previous methods, and notably two orders of magnitude fewer samples than OpenCLIP. We believe that further investment towards improving the video pretraining data distribution could lead to substantial gains in downstream image and video tasks.

E.4 Masking Strategy

An important component of the V-JEPA pretraining strategy is the 3D clip masking strategy. In this section, we detail 26 ablation experiments exploring different masks. For all the experiments, we pretrain a ViT-B/16 pretrained on K400. Figure 8 presents a summary of those results.

Figure 8c shows the effect of changing the spatial and temporal masking ratio. Figure 8b ablates the number of sampled blocks used to construct the masks given a fixed effective masking ratio of 90%. Finally, in Figure 8a we

表 12. V-JEPA、VideoMAE 的线性与注意力探测。适配 K400（$16 \times 5 \times 3$）、SSv2（$16 \times 2 \times 2$）时比较线性（Lin.）和注意力（Att.）探测，两者都受益于非线性注意力探测。

		K400		SSv2	
方法	**架构**	Lin.	Att.	Lin.	Att.
VideoMAE	ViT-L/16	52.5	77.8	41.3	61.2
V-JEPA	ViT-L/16	56.7	**80.8**	50.1	**69.5**

表 13. DINOv2、OpenCLIP 线性（Lin.）与注意力（Att.）适配评估。图像基线也受益于注意力探测。灰色结果来自 Oquab 等（2023）的线性评估。

		K400		SSv2		IN1K		Place205		iNat21	
方法	**架构**	Lin.	Att.	Lin.	Att.	Lin.	Att.	Lin.	Att.	Lin.	Att.
DINOv2	ViT-g/14	78.4	83.4	38.3	50.0	86.5	86.2	67.5	68.4	85.7	88.8
OpenCLIP	ViT-G/14	78.3	81.8	35.8	34.8	86.2	85.3	69.8	70.2	76.0	83.6

单片段与多片段。表 14 在 K400 动作分类中改变评估时间覆盖：VideoMAE、V-JEPA 的注意力探测器接收一个随机片段特征图，或八个随机片段的拼接特征图。多片段将视频均分八段，每段随机采一片段。单片段平均 $\approx$ 2秒，八片段 $\approx$ 16 秒。编码器分别处理各片段，其特征图在探测器输入处拼接。

覆盖从每视频 1 增至 8 片段，可改善 V-JEPA、VideoMAE 的 K400 动作分类，因此默认采用多片段注意力探测。

E.2 微调

表 15 分别在 K400、SSv2 微调 V-JEPA，与采用 ViT-L/16 或 ViT-H/16 的 VideoMAEv2（Wang 等，2023a）、VideoMAE（Tong 等，2022）、MVD（Wang 等，2023b）比较。V-JEPA 微调表现具有竞争力。ViT-H/16 在 SSv2 比 VideoMAE 高 1.2%，比 VideoMAEv2 高 +0.3%，K400 表现相近；SSv2 也与 MVD 相近。MVD 在 K400 最佳，但它还用 ImageNet1K 图像训练，表内其他方法只用视频。MVD 还需预先训练教师编码器，因此预训练处理样本显著更多。

E.3 预训练样本效率

比较先进图像、视频模型的预训练样本效率，即网络处理图像或视频片段总数，多轮训练时超过数据集大小。V-JEPA 结果使用的样本比此前方法少一个数量级，比 OpenCLIP 少两个数量级。进一步改善视频预训练数据分布，可能大幅提升下游图像与视频性能。

E.4 掩码策略

三维片段遮蔽是重要预训练组成。本节详述 26 项不同掩码消融，均在 K400 预训练 ViT-B/16。图 8 汇总结果。

图 8c 改变时空比例，图 8b 固定有效遮蔽 90% 消融块数；图 8a 则

Table 14 Temporal Coverage on Kinetics-400. We evaluate the effect of temporal coverage on K400. We train an attentive probe on K400 using either 1 clip (≈ 2 seconds of a video) or 8 clips (≈ 16 seconds of a video). To sample N clips, we first divide a video in N equal-length temporal segments and sample one clip at random per segment. The video encoder processes each clip in parallel and all the encoder output tokens are concatenated at the input of the attentive probe. Increasing the temporal coverage from 1 clip per video to 8 clips significantly improves the performance for both our VideoMAE baseline and V-JEPA.

Method	Arch.	1 Clip	8 Clips
VideoMAE	ViT-L/16	69.4	77.8
V-JEPA	ViT-L/16	73.7	80.9

Table 15 Finetuning results. We evaluate a V-JEPA model with the finetuning protocol on the K400 and SSv2 datasets using 16 frames per clip and multi-view fusion (5×3 or 2×3) for inference. The #**Samples Seen** entry corresponds to the number of video clips processed during pretraining, which is larger than the size of the pretraining dataset for multi-epoch training. We compare V-JEPA with different video self-supervised learning approaches. We report the VideoMAEv2 results without instruction-turning for consistency with the other approaches. V-JEPA obtains competitive performance using the finetuning protocol.

Method	Arch.	Pretraining Data	#Samples Seen	K400 (16×5×3)	SSv2 (16×2×3)
VideoMAEv1	ViT-L/16	K400|SSv2	380M|410M	85.4	74.3
	ViT-H/16	K400|SSv2	380M|410M	86.6	74.8
VideoMAEv2	ViT-H/16	Un.Hybrid	1600M	86.9	76.8
MVD	ViT-L/16	K400+IN1K	2400M	86.4	76.7
	ViT-H/16	K400+IN1K	2400M	**87.2**	**77.3**
V-JEPA	ViT-L/16	VideoMix2M	270M	85.6	75.1
	ViT-H/16	VideoMix2M	270M	86.6	77.0

examine our multi-masking strategy and find that sampling two masks for each clip (long-range and short-range) to be more effective than sampling just a single mask for each clip.

In Figure 8c, we explore different average spatial and temporal masking ratio, i.e. the spatial/temporal ratio of the area that is covered by a mask on average for a clip. Recall that each mask is constructed by sampling several (possibly overlapping) blocks and taking their union. We change the average spatial or temporal masking ratio by changing a block spatial or temporal size, as well as the overall number of blocks. We found that low spatial or temporal coverage results in a trivial prediction task, which degrades downstream performance. Based on those results, we sample masks that remove roughly 90% of the frame and extend along the entire temporal dimension of the clip by default.

In Figure 8b , we explore different block size given an effective spatial masking ratio of 90% and temporal ratio of 100%. We keep the masking ratio approximately constant by changing the block size and the number of block at the same time. We find that sampling several blocks to perform better than sampling a single large block. Figure 9 visually illustrates the effect of sampling several smaller blocks to construct a mask.

In Figure 8a, we explore the effect of sampling various number of masks per samples. We find that sampling two masks for each clip, with different spatial block sizes for each, to be more effective than sampling just a single mask. We hypothesize that this masking strategy induces complementary tasks. In our experiment, we use this as our default masks sampling.

表 14. Kinetics-400 时间覆盖。注意力探测器使用 1 片段（≈ 2 秒）或 8 片段（≈ 16 秒）。抽取 N 片段时，先将视频均分 N 段，每段随机采一片段。编码器并行处理，全部输出令牌在探测器输入处拼接。覆盖由 1 增至 8 片段，VideoMAE 与 V-JEPA 均明显提高。

方法	架构	1 个片段	8 个片段
VideoMAE	ViT-L/16	69.4	77.8
V-JEPA	ViT-L/16	73.7	80.9

表 15. 微调结果。K400、SSv2 上每片段 16 帧，推理多视图融合（5×3 或 2×3）。已见样本数是预训练处理片段总数，多轮训练时大于数据集规模。比较各视频自监督方法；为保持一致，VideoMAEv2 不使用指令调优的结果。V-JEPA 微调表现具有竞争力。

方法	架构	预训练数据	已见样本数	K400 (16×5×3)	SSv2 (16×2×3)
VideoMAEv1	ViT-L/16	K400|SSv2	380M|410M	85.4	74.3
	ViT-H/16	K400|SSv2	380M|410M	86.6	74.8
VideoMAEv2	ViT-H/16	无标签混合	1600M	86.9	76.8
MVD	ViT-L/16	K400+IN1K	2400M	86.4	76.7
	ViT-H/16	K400+IN1K	2400M	**87.2**	**77.3**
V-JEPA	ViT-L/16	VideoMix2M	270M	85.6	75.1
	ViT-H/16	VideoMix2M	270M	86.6	77.0

研究多掩码策略，发现每片段长范围与短范围两个掩码，比单个掩码更有效。

图 8c 探索平均空间和时间掩码比例，即片段被覆盖面积的平均比例。每个掩码由多个可能重叠的块取并集形成。改变块的空间／时间大小及总数，即可改变比例。覆盖过低会使预测任务平凡，降低下游性能。因此默认掩码移除约 90% 帧面积，贯穿片段全部时间。

图 8b 固定有效空间遮蔽 90%、时间 100%，改变块大小，同时调整块数保持比例约不变。多个块优于单个大块。图 9 展示多个小块构建掩码的效果。

图 8a 改变每样本掩码数。每片段使用两种空间块大小的两个掩码，优于单个掩码。我们推测它产生互补任务，将其用作默认抽样。

Table 16 Sample efficiency. We compare the sample efficiency of pretraining various state-of-the-art image and video models. The #**Samples Seen** entry corresponds to the number of samples (image or video clips) processed by the network during pretraining, which is larger than the size of the pretraining dataset for multi-epoch training. The V-JEPA results in this paper are obtained while processing an order of magnitude fewer samples than previous methods.

Method	Arch.	Data	#Samples Seen
OpenCLIP	ViT-G/14	LAION-2B	39000M
DINOv2	ViT-g/14	LVD 142M	1900M
VideoMAEv2	ViT-g/14	UnlabeledHybrid	1600M
V-JEPA	ViT-H/16_{384}	VideoMix2M	210M

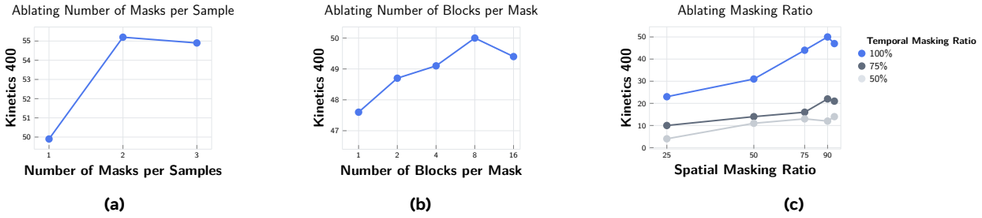
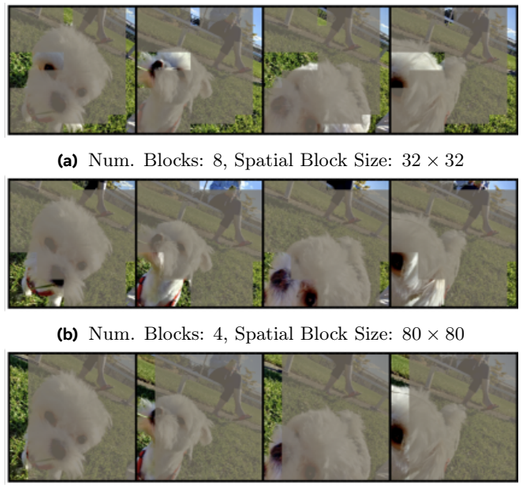


Figure 8 Masking Strategy Ablation. Evaluating a linear probe on a ViT-B/16 pretrained with V-JEPA on K400 under various 3D Multi-Block masking settings. We examine the impact of **(a)** sampling several masks per video, **(b)** varying the number of blocks in a mask, and **(c)** varying the average spatial and temporal masking ratio. A temporal masking ratio of 100% extends the spatial mask across all the frames in the clip. We find it important to maintain a high spatial and temporal masking ratio during pretraining.



(a) Num. Blocks: 8, Spatial Block Size: 32×32

(b) Num. Blocks: 4, Spatial Block Size: 80×80

(c) Num. Blocks: 2, Spatial Block Size: 160×160

Figure 9 Illustration of mask with number of blocks and block size. Each mask is constructed by sampling several (possibly overlapping) blocks and taking their union.

表 16. 样本效率。比较先进图像、视频模型的预训练。已见样本数是网络处理图像或片段总数，多轮时大于数据集大小。本文 V-JEPA 的处理样本比此前方法少一个数量级。

Method	Arch.	Data	#Samples Seen
OpenCLIP	ViT-G/14	LAION-2B	39000M
DINOv2	ViT-g/14	LVD 142M	1900M
VideoMAEv2	ViT-g/14	UnlabeledHybrid	1600M
V-JEPA	ViT-H/16_{384}	VideoMix2M	210M

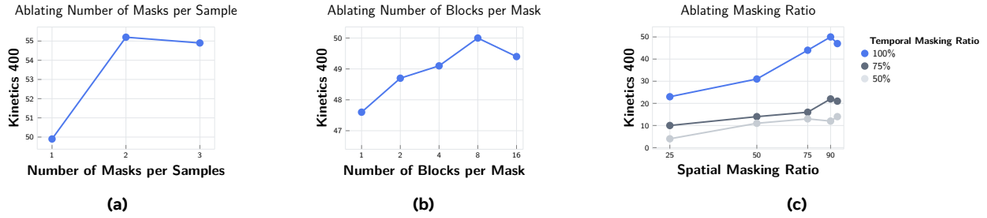
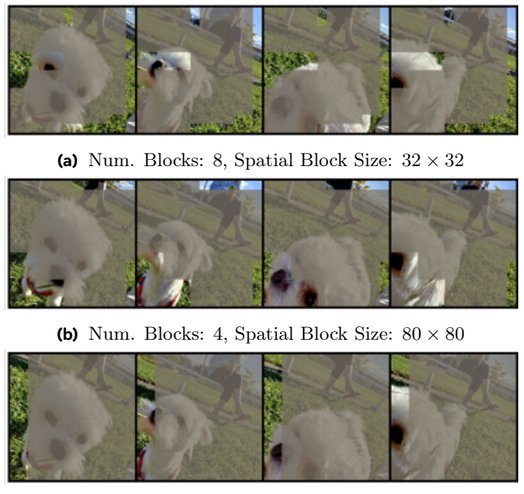


图 8. 掩码策略消融。ViT-B/16 在 K400 以不同三维多块设置预训练，线性探测评估：（a）每视频多个掩码；（b）每掩码块数；（c）平均时空比例。100% 时间比例意味着空间掩码贯穿全部帧。预训练应维持高空间和时间比例。



(a) Num. Blocks: 8, Spatial Block Size: 32×32

(b) Num. Blocks: 4, Spatial Block Size: 80×80

160×160

图 9. 块数和大小对掩码的影响。每个掩码由多个可能重叠的随机块取并集构成。